Linear Algebra 2

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Abstract

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6 Lecture 6

How to relate different vector spaces.

6.1 Linear maps

Definition 6.1

Linear map

Let V and W be vector spaces over a field $\mathbb{F}$. A map $T : V \to W$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$T(u+v) = T(u) + T(v), \quad T(\alpha v) = \alpha T(v).$$

Additivity and homogeneity

- Additivity: $T(u+v) = T(u) + T(v)$ for all $u, v \in V$.
- Homogeneity: $T(\alpha v) = \alpha T(v)$ for all $\alpha \in \mathbb{F}, v \in V$.

Notation

- $\mathcal{L}(V, W) =$ the space of linear maps $T : V \to W$.
- $\mathcal{L}(V) =$ the space of linear maps $T : V \to V$.

6.2 Examples

6.2.1 The zero linear map

$$\begin{aligned} 0 \colon V &\to W, \\ v &\mapsto 0. \end{aligned}$$

Remark

$T(0_V) = 0_W$.

$$\begin{aligned} 0_V &= 0_{\mathbb{F}} v, \\ T(0_V) &= T(0_{\mathbb{F}} v) = 0_{\mathbb{F}} T(v) = 0_W. \end{aligned}$$

6.2.2 The identity linear map

For any $v \in V$,

$$\begin{aligned} I \colon V &\to V, \\ v &\mapsto v. \end{aligned}$$

6.2.3 Differentiation operator

First recall the definition of a polynomial space over $\mathbb{R}$.

$$\mathcal{P}(\mathbb{R}) = \{\text{polynomials } a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_i \in \mathbb{R}, n \in \mathbb{N}\}.$$

Then define the differentiation operator.

$$\begin{aligned} D \colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \psi'. \end{aligned}$$

For $\psi \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \psi &= a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \\ \psi' &= a_1 + 2a_2 x + \cdots + n a_n x^{n-1}. \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} (p + q)' &= p' + q', \\ (\alpha p)' &= \alpha p'. \end{aligned}$$

The differentiation operator is linear.

6.2.4 Integration operator

Define the integration operator.

$$\begin{aligned} \int : \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \int \psi \, dx. \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \int (p+q) &= \int p + \int q, \\ \int (\alpha p) &= \alpha \int p. \end{aligned}$$

6.2.5 Multiplication by a fixed polynomial

Define the multiplication by a fixed polynomial operator.

$$\begin{aligned} T \colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto q\psi. \end{aligned}$$

So

$$(Tp)(x) = q(x)p(x).$$

Check that T is linear.

6.2.6 Backward shift operator

Define the backward shift operator.

$$\begin{aligned} T \colon \mathbb{F}^\infty &\to \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (x_2, x_3, \dots). \end{aligned}$$

Check that T is linear.

6.2.7 Forward shift operator

Define the forward shift operator.

$$\begin{aligned} T \colon \mathbb{F}^\infty &\to \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (0, x_1, x_2, \dots). \end{aligned}$$

Check that T is linear.

6.2.8 Linear map and matrices

$$\begin{aligned} &T \colon \mathbb{F}^n \to \mathbb{F}^m, \\ &\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}. \end{aligned}$$

T should be a matrix.

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{(m\times n)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$m \times n$ matrix: rows $\times$ columns.

$$T\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; T\begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; \dots, \; T\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Columns!
So, because the lemma below shows that a linear map is completely determined by its action on the basis vectors of the domain. Any linear map can be uniquely represented by a matrix with respect to the chosen bases.

6.3 A linear map is determined by its values on a basis

Lemma 6.1

A linear map $T\colon V \to W$ is uniquely determined by what it does on a basis of V.

Example

For a basis $\{v_1, \dots, v_n\}$ of V, let

$$T(v_i) = w_i \in W \quad \text{for } i = 1, 2, \dots, n.$$

$$\begin{aligned} T(v_1) &= w_1, \\ T(v_2) &= w_2, \\ &\vdots \\ T(v_n) &= w_n. \end{aligned}$$

Then T is uniquely determined by the images of the basis vectors.

$$\begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix}$$

determines T uniquely. For $v \in V$,

$$\begin{aligned} Tv &= T\left(a_1v_1 + a_2v_2 + \cdots + a_nv_n\right) \\ &= a_1T(v_1) + a_2T(v_2) + \cdots + a_nT(v_n) \\ &= a_1w_1 + a_2w_2 + \cdots + a_nw_n. \end{aligned}$$

Check that T is a linear map.

6.3.1 $\mathcal{L}(V, W)$

Let $S, T \in \mathcal{L}(V, W)$.

$$\begin{aligned} (S+T)(v) &= S(v) + T(v) \quad && \text{for all } v \in V, \\ (\lambda S)(v) &= \lambda S(v) && \text{for all } v \in V \text{ and } \lambda \in \mathbb{F}. \end{aligned}$$

Claim

(Need to check)

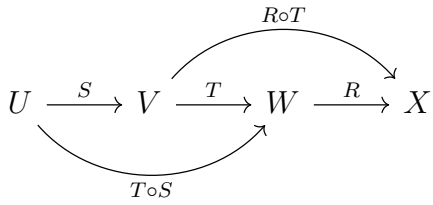
$$S + T, \lambda S \quad \text{are linear maps.}$$

That makes $\mathcal{L}(V, W)$ a vector space over $\mathbb{F}$.
Similarly, $\mathcal{L}(V)$ is a vector space over $\mathbb{F}$ for any vector space V.

6.3.2 Compositions of linear maps

Case 1

Let U, V, W be vector spaces over $\mathbb{F}$.



For all $u \in U$,

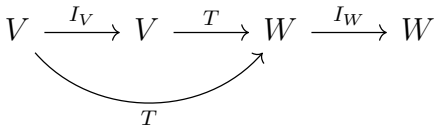
$$\begin{aligned} (T \circ S)(u) &= T(S(u)), \\ (R \circ (T \circ S))(u) &= R(T(S(u))), \\ ((R \circ T) \circ S)(u) &= (R \circ T)(S(u)) = R(T(S(u))). \end{aligned}$$

Therefore,

$$R \circ (T \circ S) = (R \circ T) \circ S.$$

Case 2

For $T \colon V \to W$.



So

$$\begin{aligned} T \circ I_V &= T, \\ I_W \circ T &= T. \end{aligned}$$

For $S, T \colon V \to V$, the compositions $S \circ T$ and $T \circ S$ need not agree.

Example

For $V = \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} T\colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto p'(x). \end{aligned}$$

$$\begin{aligned} S\colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto xp(x). \end{aligned}$$

$T(p(x)) = p'(x), S(p(x)) = xp(x)$.

$$\begin{aligned} (S \circ T)(p(x)) &= S(p'(x)) = xp'(x), \\ (T \circ S)(p(x)) &= T(xp(x)) = p(x) + xp'(x). \end{aligned}$$

6.4 Kernel and Image of a linear map

Definition 6.2

Kernel and Image Spaces

Let $T\colon V \to W$ be a linear map.

$$\begin{aligned} \ker T &= \{v \in V \mid T(v) = 0_W\} && \text{kernel of } T \text{ (null space of } T), \\ \operatorname{Im} T &= \{T(v) \mid v \in V\} && \text{image of } T \text{ (range of } T). \end{aligned}$$

Example

Zero map

For the zero map,

$$\begin{aligned} 0\colon V &\to W \\ v &\mapsto 0. \end{aligned}$$

We can immediately see that the range of the zero map is just the zero vector in W, and the null space is the entire space V.

$$\begin{aligned} \operatorname{Im}(0) &= \{0\}, \\ \ker(0) &= V. \end{aligned}$$

Example

Differentiation map

For the differentiation map,

$$D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$$
$$\gamma \mapsto \gamma'.$$

We can immediately see that the range of the differentiation map is the set of all derivatives of functions in $\mathcal{P}(\mathbb{R})$, and the null space is the set of all constant polynomials.

$$\operatorname{Im}(D) = \{\gamma' \mid \gamma \in \mathcal{P}(\mathbb{R})\},$$
$$\ker(D) = \{\text{constant polynomials}\}.$$

7 Lecture 7

7.1 Properties of Linear Maps

Let V and W be vector spaces over the same field $\mathbb{F}$. Let $T\colon V \to W$ be a linear map. For all $v_1, v_2 \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{cases} T(v_1 + v_2) = T(v_1) + T(v_2), \\ T(\alpha v_1) = \alpha T(v_1), \end{cases} \iff T(u + \alpha v) = T(u) + T(\alpha v) = T(u) + \alpha T(v).$$

7.2 Injective and Surjective Maps

7.2.1 Kernel and Image Definition

Definition 7.1

The kernel of T

Let $T\colon V \to W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\}.$$

Definition 7.2

The image of T

Let $T\colon V \to W$ be a linear map.

$$\operatorname{Im} T = \{T(v) \mid v \in V\}.$$

Theorem 7.1

The kernel and image of a linear map T are vector subspaces of V and W, respectively.

$$\ker T \subseteq V,$$
$$\operatorname{Im} T \subseteq W.$$

Proof. Since $T(0_V) = 0_W$, the kernel contains 0_V.
If $v_1, v_2 \in \ker T$ and $\lambda \in \mathbb{F}$, then

$$T(v_1 + \lambda v_2) = T(v_1) + \lambda T(v_2) = 0_W,$$

so $v_1 + \lambda v_2 \in \ker T$.
Thus $\ker T$ is a subspace of V. □

Proof. If $w_1, w_2 \in \operatorname{Im} T$, write

$$w_1 = T(v_1),$$
$$w_2 = T(v_2)$$

for some $v_1, v_2 \in V$. For $\lambda \in \mathbb{F}$,

$$w_1 + \lambda w_2 = T(v_1) + \lambda T(v_2) = T(v_1 + \lambda v_2) \in \operatorname{Im} T.$$

Thus $\operatorname{Im} T$ is a subspace of W. □

7.2.2 Injective

Definition 7.3

The linear map $T\colon V \to W$ is called *injective* (or one-to-one) if $\ker T = \{0_V\}$.

For $T\colon V \to W$, injectivity is equivalent to

$$T(v_1) = T(v_2) \implies v_1 = v_2.$$

Indeed, if $T(v_1) = T(v_2)$, then

$$T(v_1 - v_2) = 0_W,$$

so

$$v_1 - v_2 \in \ker T = \{0_V\},$$

and hence

$$v_1 = v_2.$$

Conversely, if T is injective and $v \in \ker T$, then

$$T(v) = T(0_V),$$

so $v = 0_V$.

7.2.3 Surjective

Definition 7.4

The linear map $T\colon V \to W$ is called *surjective* (or onto) if $\operatorname{Im} T = W$.

If T is surjective, then by definition $\operatorname{Im} T = W$.
Equivalently, for every $w \in W$ there exists $v \in V$ such that

$$w = T(v).$$

Example

The derivative map

$$D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$$
$$p \mapsto p'.$$

has

$$\ker D = \{p \in \mathcal{P}(\mathbb{R}) \mid D(p) = 0\},$$
$$\operatorname{Im} D = \{D(p) \mid p \in \mathcal{P}(\mathbb{R})\}.$$

D is surjective but not injective.

Example

The backward shift b and forward shift f on $\mathbb{F}^\infty$ are defined by

$$\mathbb{F}^\infty \underset{f}{\overset{b}{\rightleftarrows}} \mathbb{F}^\infty$$

$$b(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots),$$
$$f(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots).$$

The kernels and images of the backward and forward shifts are given by

$$\ker b = \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_2 = x_3 = \cdots = 0\},$$
$$\operatorname{Im} b = \mathbb{F}^\infty,$$
$$\ker f = \{0\},$$
$$\operatorname{Im} f = \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_1 = 0\}.$$

The backward shift is surjective but not injective, while the forward shift is injective but not surjective.

Example

$$T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R}),$$
$$p(x) \mapsto xp(x).$$
$$\ker T = \{p(x) \in \mathcal{P}(\mathbb{R}) \mid xp(x) = 0\} = \{0\}.$$
$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \quad a_i \in \mathbb{R}.$$
$$xp(x) = a_0 x + a_1 x^2 + a_2 x^3 + \cdots + a_n x^{n+1}.$$
$$\operatorname{Im} T = \{xp(x) \mid p(x) \in \mathcal{P}(\mathbb{R})\} \neq \mathcal{P}(\mathbb{R}).$$

T is injective but not surjective.

7.3 The Fundamental Theorem of Linear maps

Theorem 7.2

The fundamental theorem of linear maps (also known as the rank-nullity or rank theorem)

If $T\colon V \to W$ is a linear map, and the dimension of V is finite, then

$$\dim V = \underbrace{\dim \ker T}_{\text{nullity}} + \underbrace{\dim \operatorname{Im} T}_{\text{rank}}.$$

Proof. Since $\ker T \subseteq V$, let $\{u_1, \dots, u_n\}$ be a basis for $\ker T$. Extend it to a basis for V.

$$\{u_1, \dots, u_n, v_1, \dots, v_r\}$$

Thus

$$\dim V = n + r, \quad \dim \ker T = n.$$

We want to show that $\dim \operatorname{Im} T = r$, so we claim that $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$, need to be proven.

Claim: $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

1. $\{T(v_1), \dots, T(v_r)\}$ spans $\operatorname{Im} T$. Since $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V, any $v \in V$ can be written as

   $$v = a_1u_1 + \cdots + a_nu_n + b_1v_1 + \cdots + b_rv_r.$$

   Applying T, we get

   $$T(v) = b_1T(v_1) + \cdots + b_rT(v_r),$$

   since $T(u_i) = 0$ for all i. Therefore, $\{T(v_1), \dots, T(v_r)\}$ spans $\operatorname{Im} T$.

2. $\{T(v_1), \dots, T(v_r)\}$ is linearly independent. Suppose

   $$c_1T(v_1) + \cdots + c_rT(v_r) = 0.$$

   Then

   $$T(c_1v_1 + \cdots + c_rv_r) = 0.$$

   Therefore, $c_1v_1 + \cdots + c_rv_r \in \ker T$. But $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V, so $c_1v_1 + \cdots + c_rv_r$ can be written uniquely as a linear combination of the basis vectors:

   $$c_1v_1 + \cdots + c_rv_r = a_1u_1 + \cdots + a_nu_n.$$

   Hence

   $$c_1v_1 + \cdots + c_rv_r - (a_1u_1 + \cdots + a_nu_n) = 0.$$

   Since the basis vectors are linearly independent, we must have

   $$c_1 = \cdots = c_r = 0.$$

   This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim. Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

$$\dim \operatorname{Im} T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

$$a_1 = \cdots = a_n = 0, \quad c_1 = \cdots = c_r = 0.$$

This proves that $\{T(v_1), \ldots, T(v_r)\}$ is linearly independent, completing the proof of the claim.

Therefore, $\{T(v_1), \ldots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

$$\dim \operatorname{Im} T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

□

7.4 Corollaries of the Rank Theorem

Let $T\colon V \to W$ with

1. $\dim V < \infty$
2. $\dim W < \infty$.

7.4.1 Injective Maps

Corollary 7.3

If T is injective, then

$$\dim V = \dim \operatorname{Im} T \leq \dim W,$$

because $\ker T = \{0_V\}$.

So if $\dim V > \dim W$, no injective map $T\colon V \to W$ exists.

7.4.2 Surjective Maps

Corollary 7.4

If T is surjective, then

$$\dim W = \dim \operatorname{Im} T \leq \dim V,$$

because $\operatorname{Im} T = W$.

So if $\dim V < \dim W$, no surjective map $T\colon V \to W$ exists.

7.4.3 Bijective Maps

Corollary 7.5

If T is bijective (injective and surjective), then

$$\dim V = \dim W.$$

This follows directly from the corollaries for injective and surjective maps.

Proof. Since T is injective,

$$\dim V = \dim \operatorname{Im} T \leq \dim W.$$

Since T is surjective,

$$\dim W = \dim \operatorname{Im} T \leq \dim V.$$

Therefore,

$$\dim V = \dim W.$$

□

8 Lecture 8

8.1 Review

For $T\colon V \to W$,

$$\ker T = \{v \in V \mid T(v) = 0_W\} \subseteq V,$$
$$\operatorname{Im} T = \{T(v) \mid v \in V\} \subseteq W.$$

- T is injective if $\ker T = \{0_V\}$.
- T is surjective if $\operatorname{Im} T = W$.

8.2 Invertible Maps

Definition 8.1

A linear map $T\colon V \to W$ is *invertible* if there exists a linear map $S\colon W \to V$ such that

$$S \circ T = I_V \quad \text{and} \quad T \circ S = I_W.$$

S is an *inverse* of T.

Theorem 8.1

An invertible linear map $T\colon V \to W$ has a unique inverse.

Proof. Suppose T has two inverses S_1 and S_2. Then

$$T \circ S_1 = I_W \quad \text{and} \quad T \circ S_2 = I_W,$$

$$S_1 \circ T = I_V \quad \text{and} \quad S_2 \circ T = I_V.$$

Therefore,

$$S_1 = S_1 \circ I_W = S_1 \circ (T \circ S_2) = (S_1 \circ T) \circ S_2 = I_V \circ S_2 = S_2.$$

Hence, the inverse of T is unique. □

Theorem 8.2

If $T\colon V \to W$ is a linear map, then

$$T \text{ is invertible} \iff T \text{ is both injective and surjective.}$$

Proof. Proof in both directions.

($\Rightarrow$) If T is invertible, then it is both injective and surjective

Show T is injective
Let $v \in V$ such that $T(v) = 0_W$, then

$$v = T^{-1}T(v) = T^{-1}(0_W) = 0_V.$$

Thus

$$\ker T = \{0_V\} \implies T \text{ is injective.}$$

Show T is surjective
Let $w \in W$, then

$$w = T(T^{-1}(w)) \implies w \in \text{im } T.$$

So

$$W \subseteq \text{im } T \implies \text{im } T = W \implies T \text{ is surjective.}$$

($\Leftarrow$) If T is both injective and surjective, then it is invertible

Since T is surjective, $\forall w \in W$, $\exists v \in V$ such that $T(v) = w$.
Define

$$\begin{aligned} T^{-1} &: W \to V, \\ w &\mapsto v. \end{aligned}$$

This is well-defined because T is injective.
Then

$$T^{-1} \circ T(v) = v \quad \text{and} \quad T \circ T^{-1}(w) = T(v) = w$$

so

$$T^{-1} \circ T = I_V \quad \text{and} \quad T \circ T^{-1} = I_W$$

so T is invertible. □

Linearity of the Inverse

Claim T^{-1} is linear. Let $w_1, w_2 \in W$ and $\alpha, \beta \in \mathbb{F}$. Then

$$\begin{aligned} T^{-1}(\alpha w_1 + \beta w_2) &= T^{-1}(\alpha T(v_1) + \beta T(v_2)) \\ &= T^{-1}(T(\alpha v_1 + \beta v_2)) \\ &= \alpha v_1 + \beta v_2 \\ &= \alpha T^{-1}(w_1) + \beta T^{-1}(w_2) \end{aligned}$$

where $v_1 = T^{-1}(w_1)$ and $v_2 = T^{-1}(w_2)$. Hence T^{-1} is linear.

Example

Consider the forward and backward linear maps

$$b\colon \mathbb{F}^\infty \to \mathbb{F}^\infty,$$
$$(x_1, x_2, \dots) \mapsto (x_2, x_3, \dots).$$

$$f\colon \mathbb{F}^\infty \to \mathbb{F}^\infty,$$
$$(x_1, x_2, \dots) \mapsto (0, x_1, x_2, \dots).$$

b is surjective but not injective, f is injective but not surjective $\implies$ neither b nor f is invertible.

$$b \circ f = I_{\mathbb{F}^\infty} \quad \text{but} \quad f \circ b \neq I_{\mathbb{F}^\infty}.$$

This cannot happen for finite-dimensional vector spaces.

Theorem 8.3

Let $T\colon V \to W$ be a linear map with $\dim V = \dim W < \infty$. Then

$$T \text{ is injective} \iff T \text{ is surjective} \iff T \text{ is invertible.}$$

Proof. By the rank-nullity theorem, we have

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim W.$$

$$\begin{aligned} T \text{ injective} &\implies \dim \ker T = 0 \\ &\implies \dim \operatorname{Im} T = \dim V = \dim W \\ &\implies T \text{ is surjective.} \end{aligned}$$

$$\begin{aligned} T \text{ surjective} &\implies \dim \operatorname{Im} T = \dim W = \dim V \\ &\implies \dim \ker T = 0 \\ &\implies T \text{ is injective.} \end{aligned}$$

Hence T is invertible. □

Theorem 8.4

Let $T\colon V \to W$ and $S\colon W \to V$ be linear maps with $\dim V = \dim W < \infty$. Then

$$S \circ T = I_V \iff T \circ S = I_W.$$

Proof. ($\Rightarrow$) Assume $S \circ T = I_V$. Let $v \in \ker T$. Then

$$v = I_V(v) = (S \circ T)(v) = S(T(v)) = S(0_W) = 0_V.$$

Thus

$$\ker T = \{0_V\}$$

so T is injective.
Since $\dim V = \dim W < \infty$, T is invertible let $T^{-1}\colon W \to V$ be its inverse.
Then

$$\begin{aligned} S = S \circ I_W &= S \circ (T \circ T^{-1}) \\ &= (S \circ T) \circ T^{-1} \\ &= I_V \circ T^{-1} = T^{-1}. \end{aligned}$$

Hence

$$T \circ S = T \circ T^{-1} = I_W.$$

($\Leftarrow$) Repeat the proof with T and S swapped, and V and W swapped as well. □

8.3 Isomorphism

Definition 8.2

Let $T\colon V \to W$. T is called an *isomorphism* if T is invertible. In this case we write

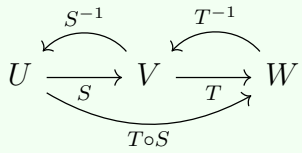
$$V \cong W,$$

and V and W are called *isomorphic*.

Theorem 8.5

Composition of isomorphisms is an isomorphism.
If $S\colon U \to V$ and $T\colon V \to W$ are isomorphisms, then $T \circ S\colon U \to W$ is an isomorphism.

S, T are isomorphisms.



$T \circ S\colon U \to W$ is an isomorphism.

Proof. Since S and T are isomorphisms, there exist

$$S^{-1}\colon V \to U, \qquad T^{-1}\colon W \to V.$$

Claim

$$S^{-1} \circ T^{-1} = (T \circ S)^{-1}.$$

Indeed,

$$\begin{aligned} T \circ S \circ (S^{-1} \circ T^{-1}) &= T \circ \underbrace{(S \circ S^{-1})}_{I_V} \circ T^{-1} \\ &= T \circ I_V \circ T^{-1} \\ &= T \circ T^{-1} \\ &= I_W \end{aligned}$$

and

$$\begin{aligned} (S^{-1} \circ T^{-1}) \circ (T \circ S) &= S^{-1} \circ \underbrace{(T^{-1} \circ T)}_{I_V} \circ S \\ &= S^{-1} \circ I_V \circ S \\ &= S^{-1} \circ S \\ &= I_U \end{aligned}$$

Thus $T \circ S$ is invertible with inverse $S^{-1} \circ T^{-1}$:

$$(T \circ S)^{-1} = S^{-1} \circ T^{-1}.$$

Therefore $T \circ S$ is an isomorphism. □

Lemma 8.6

Let V be an n-dimensional vector space over $\mathbb{F}$, and let $\{v_1, \ldots, v_n\}$ be a basis of V. Then there exists a unique isomorphism

$$\psi \colon \mathbb{F}^n \to V$$

such that

$$\psi(e_i) = v_i \qquad (i = 1, \ldots, n).$$

Proof. Define linear map

$$\psi(\alpha_1, \ldots, \alpha_n) = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

Since

$$span\{v_1, \ldots, v_n\} = V \implies \psi \text{ is surjective}$$

Since

$$\{v_1, \ldots, v_n\} \text{ is linearly independent} \implies \ker \psi = \{0\}$$

So ψ is injective. Hence ψ is an isomorphism. □

Theorem 8.7

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$. Then

$$V \cong W \iff \dim V = \dim W.$$

Proof. ($\Rightarrow$) Suppose $V \cong W$, then there exists an invertible linear map $T \colon V \to W$, which is injective and surjective.

$$\begin{aligned} \dim V &= \dim \ker T + \dim \operatorname{Im} T \\ &= 0 + \dim W \\ &= \dim W. \end{aligned}$$

($\Leftarrow$) Suppose $\dim V = \dim W = n$. Choose basis for V and W

$$v_1, \ldots, v_n$$
$$w_1, \ldots, w_n$$

By the lemma above, there are isomorphisms

$$\begin{aligned} \psi \colon \mathbb{F}^n &\to V \\ e_i &\mapsto v_i, \end{aligned}$$

and

$$\begin{aligned} \chi \colon \mathbb{F}^n &\to W \\ e_i &\mapsto w_i, \end{aligned}$$

for $i = 1, \ldots, n$. Then

$$\chi \circ \psi^{-1} \colon V \to W$$

is a composition of isomorphisms, hence an isomorphism.
Therefore

$$V \cong W$$

□

9 Lecture 9

From linear map to the matrices.

9.1 The space of matrices

Definition 9.1

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$.

$$\mathbb{F}^{m,n} = \{m \times n \text{ matrices over } \mathbb{F}\}.$$

$(m \times n)$ matrices m #rows n #columns.

$\mathbb{F}^{m,n}$ is a vector space over $\mathbb{F}$.

$$[A_{ij}]_{\substack{i=1,\dots,m \\ j=1,\dots,n}} = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}$$

$$[A_{ij}] + [B_{ij}] = [A_{ij} + B_{ij}]$$
$$\lambda[A_{ij}] = [\lambda A_{ij}]$$

Basis of $\mathbb{F}^{m,n}$ consists of matrices

$$E_{\substack{i=1,\dots,m \\ j=1,\dots,n}}$$

With a single entry 1 at the (i,j)-th position and the rest 0.

$$E_{ij} = \begin{array}{c} \\ \\ i \\ \\ \\ \end{array} \overset{\begin{array}{ccccc} & & j & & \end{array}}{\begin{bmatrix} 0 & \cdots & & \cdots & 0 \\ \vdots & & & & \vdots \\ & \cdots & 1 & \cdots & \\ \vdots & & & & \vdots \\ 0 & \cdots & & \cdots & 0 \end{bmatrix}}$$

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} E_{ij}.$$

So the dimension of $\mathbb{F}^{m,n}$ is the number of different place in the matrix, which is mn.

$$\dim \mathbb{F}^{m,n} = mn$$

9.2 Matrix representation of linear maps

Definition 9.2

Matrix multiplication

$$\mathbb{F}^{m,n} \times \mathbb{F}^{n,p} \to \mathbb{F}^{m,p}$$
$$[A_{ij}] \times [B_{jk}] \mapsto [C_{ik}]$$
$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$
$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots & \vdots \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} & B_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots & \vdots \end{bmatrix}$$
$$C_{11} = A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} + \cdots + A_{1n}B_{n1}$$

Theorem 9.1

$A \in \mathbb{F}^{m,n} \sim$ linear map $\mathbb{F}^n \to \mathbb{F}^m$

$$\mathbb{F}^n \to \mathbb{F}^m$$
$$x \mapsto Ax$$
$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}$$

For vector spaces V and W

$$\dim V = n \quad \dim W = m$$

T is a linear map from V to W

$$V \xrightarrow{T} W$$

Choose basis

$$\{v_1, \ldots, v_n\} \quad \text{for } V$$
$$\{w_1, \ldots, w_m\} \quad \text{for } W$$
$$\mathbb{F}^n \xrightarrow[\psi]{\cong} V \qquad \mathbb{F}^m \xrightarrow[\psi]{\cong} W$$
$$e_i \mapsto v_i \qquad e_i \mapsto w_i.$$

The matrix representation of the linear map T with respect to the chosen bases is given by the following commutative diagram

$$\mathbb{F}^n \xrightarrow[\psi]{\cong} V \xrightarrow{T} W \xrightarrow[\psi^{-1}]{\cong} \mathbb{F}^m$$
$$\psi^{-1} \circ T \circ \psi$$

$$T(v_i) = \sum_{j=1}^{m} w_j A_{ji}$$

$$
\begin{aligned}
T\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} &= \begin{bmatrix} Tv_1 & \cdots & Tv_n \end{bmatrix} \\
&= \begin{bmatrix} w_1 & \cdots & w_m \end{bmatrix} \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
T\left(\sum_{i=1}^{n} x_i v_i\right) &= \sum_{i=1}^{n} x_i T v_i \\
&= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} A_{ji} w_j \\
&= \sum_{j=1}^{m} w_j \left(\sum_{i=1}^{n} A_{ji} x_i\right)
\end{aligned}
$$

$\mathcal{M}(T)$ matrix which represents the linear map T with respect to the chosen bases.

Definition 9.3

The coordinate vector of a vector $w \in W$ with respect to the basis $\{w_1, \ldots, w_m\}$ is defined as follows:

$$
\begin{aligned}
\psi^{-1} &\colon W \to \mathbb{F}^n \\
w = \sum_{i=1}^{m} x_i w_i &\mapsto \begin{bmatrix} x_1 & \cdots & x_m \end{bmatrix}
\end{aligned}
$$

Coordinate vector of w with respect to the basis $\{w_1, \ldots, w_m\}$.

Example

$$\mathcal{D} \colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$$

$\{1, x, x^2, x^3\}$ basis of $\mathcal{P}_3(\mathbb{R})$ $\{1, x, x^2\}$ basis of $\mathcal{P}_2(\mathbb{R})$ Question: What is the matrix representation of the differentiation operator $\mathcal{D}$ with respect to the given bases?

$$
\mathcal{D}(1, x, x^2, x^3) = (0, 1, 2x, 3x^2) = (1, x, x^2) \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}
$$

$$
[\mathcal{M}(\mathcal{D})] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}
$$

9.3 The isomorphism $\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$ and composition

Theorem 9.2

For vector spaces V and W, let $\dim V = n$, $\dim W = m$.

$$\begin{aligned} \mathcal{L}(V,W) &\to \mathbb{F}^{m\times n} \\ T &\mapsto \mathcal{M}(T) \end{aligned}$$

is a vector space isomorphism, and hence

$$\begin{aligned} \dim \mathcal{L}(V,W) &= mn \\ &= (\dim W)(\dim V) \end{aligned}$$

Proof. (From workshop $\mathcal{M}(T) = \psi^{-1} \circ T \circ \psi$)

$$\begin{aligned} \mathcal{M}(S+T) &= \psi^{-1} \circ (S+T) \circ \psi \\ &= \psi^{-1} \circ S \circ \psi + \psi^{-1} \circ T \circ \psi \\ &= \mathcal{M}(S) + \mathcal{M}(T) \end{aligned}$$

For $x \in \mathbb{F}^n$,

$$\begin{aligned} \psi^{-1} \underbrace{(S+T)(\psi(x))}_{S(\psi(x))+T(\psi(x))} &= \psi^{-1}(S(\psi(x)) + T(\psi(x))) \\ &= \psi^{-1}(S(\psi(x))) + \psi^{-1}(T(\psi(x))) \\ &= (\psi^{-1}S\psi + \psi^{-1}T\psi)(x) \end{aligned}$$

$$\begin{aligned} \mathcal{M}(\lambda T) &= \psi^{-1} \circ (\lambda T) \circ \psi \\ &= \lambda(\psi^{-1} \circ T \circ \psi) \\ &= \lambda \mathcal{M}(T) \end{aligned}$$

$\mathcal{M}$ is a linear map.

$$\begin{aligned} \mathcal{M}(T) &= \psi^{-1} \circ T \circ \psi \\ \psi \circ \mathcal{M}(T) \circ \psi^{-1} &= \underline{\psi \circ \psi^{-1}} \circ T \circ \underline{\psi \circ \psi^{-1}} \\ &= T \end{aligned}$$

□

Theorem 9.3

Composition of linear maps is represented by matrix multiplication with respect to the chosen bases.

Proof. U, V, W are vector spaces over the field $\mathbb{F}$.

$$\dim U = p, \quad \dim V = n, \quad \dim W = m.$$

Let $S\colon U \to V$ and $T\colon W \to U$ be linear maps with chosen bases for V, W, U. Then the composition $S \circ T\colon W \to V$ is represented as follows

$$W \xrightarrow{T} U \xrightarrow{S} V, \qquad W \xrightarrow{S\circ T} V$$

Chosen basis

$$U\colon u_1, \ldots, u_p$$
$$W\colon w_1, \ldots, w_m$$
$$V\colon v_1, \ldots, v_n$$

$$Su_i = \sum_{j=1}^{n} v_j S_{ji} \qquad Tw_k = \sum_{i=1}^{p} u_i T_{ik}$$

$$(S \circ T)w_k = \sum_{j=1}^{n} v_j C_{jk}$$

where

$$C_{jk} = \sum_{i=1}^{p} S_{ji} T_{ik} \Leftrightarrow \mathcal{M}(S \circ T) = \mathcal{M}(S)\mathcal{M}(T)$$

Then

$$\begin{aligned} S(Tw_k) &= S\left(\sum_{i=1}^{p} u_i T_{ik}\right) \\ &= \sum_{i=1}^{p} T_{ik} S(u_i) \\ &= \sum_{i=1}^{p} T_{ik} \left(\sum_{j=1}^{n} v_j S_{ji}\right) \\ &= \sum_{j=1}^{n} v_j \left(\sum_{i=1}^{p} S_{ji} T_{ik}\right) \\ &= \sum_{j=1}^{n} v_j C_{jk} \end{aligned}$$

□

9.4 Normal Form for Matrices

Normal form for matrices representing linear maps (cf. proof of the Rank theorem)
For linear map T

$$T\colon V \to W$$

Choose basis for $\ker T$

$$u_1, \ldots, u_k$$

Complete it to a basis for the whole space V

$$v_1, \ldots, v_r, u_1, \ldots, u_n$$

So basis for the $\operatorname{im} T \subseteq W$ is

$$Tv_1, \ldots, Tv_r$$

Complete it to a basis for the space W

$$Tv_1, \ldots, Tv_r, w_1, \ldots, w_p$$

apply the linear map T to the basis vectors of V

$$\begin{aligned} T(v_1,\dots,v_r,u_1,\dots,u_k) &= (Tv_1,\dots,Tv_r,Tu_1,\dots,Tu_k) \\ &= (Tv_1,\dots,Tv_r,0,\dots,0) \\ &= (Tv_1,\dots,Tv_r,w_1,\dots,w_p) \end{aligned}$$

Since

$$\forall i = 1,\dots,k \quad Tu_i = 0$$

So if we choose the bases for V and W as described above, the matrix representation of T with respect to these bases is, which called the *normal form* of the matrix representation of T.

$$\mathcal{M}(T) = \begin{array}{c} \begin{matrix} \overset{r}{} & & \overset{n}{} \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & \cdots & 0 & \\ \vdots & \ddots & \vdots & 0 \\ 0 & \cdots & 1 & \\ \hline & 0 & & 0 \end{array}\right] \begin{matrix} r \\ \\ \\ p \end{matrix} \end{array}$$

$r = \operatorname{rank} T = \dim \operatorname{im} T$

$$\mathcal{M}(T) = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right)$$

Remark

Careat: For a linear map

$$T\colon V \to W$$

The poblem will be much harder.

---

cf. Jordan Normal Form

10 Lecture 10

10.1 Change of Basis

$T : V \to W$
Let V and W be vector spaces over $\mathbb{F}$ and $\dim V = n$, $\dim W = m$.

$$\begin{cases} v_1,\dots,v_n \\ w_1,\dots,w_m \end{cases} \quad \text{bases for } V, W$$

Define the isomorphisms

$$\begin{array}{cc} \mathbb{F}^n \underset{\varphi}{\overset{\cong}{\to}} V & \mathbb{F}^m \underset{\psi}{\overset{\cong}{\to}} W \\ e_i \mapsto v_i & e_i \mapsto w_i \end{array}$$

We have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle\varphi}\uparrow & & \downarrow{\scriptstyle\psi^{-1}} \\ \mathbb{F}^n & \xrightarrow[m(T)]{} & \mathbb{F}^m \end{array}$$

$$m(T) = \psi^{-1} T \varphi$$

$$Tv_i = \sum_{j=1}^{m} w_j T_{ji}$$

10.2 Change of Basis for Linear Operators

10.2.1 Change of Basis Matrices

For a finite dimensional vector space V over $\mathbb{F}$ ($\dim V = n$)

$$\begin{cases} v_1, \ldots, v_n \\ v_1', \ldots, v_n' \end{cases} \quad \text{bases for } V$$

Then for $v \in V$

$$v = \sum_{i=1}^{n} a_i v_i = \sum_{i=1}^{n} a_i' v_i'$$

$$v_i = \sum_{j=1}^{n} v_j' T_{ji}$$

$$v_i' = \sum_{j=1}^{n} v_j S_{ji}$$

T_{ji}, S_{ji} are the entries of $n \times n$ matrices T & S

$$T = \begin{pmatrix} T_{11} & \cdots & T_{1n} \\ \vdots & \ddots & \vdots \\ T_{n1} & \cdots & T_{nn} \end{pmatrix} \quad S = \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \cdots & S_{nn} \end{pmatrix}$$

$$\begin{aligned} v_i &= \sum_{j=1}^{n} v_j' T_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k S_{kj} \right) T_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} S_{kj} T_{ji} \right) \end{aligned}$$

Linear independence of $\{v_1, \ldots, v_n\}$ implies that

$$\underbrace{\sum_{j=1}^{n} S_{kj} T_{ji}}_{\text{(ki) entry of } ST} = \begin{cases} 1 & k = i \\ 0 & k \neq i \end{cases}$$

For $n \times n$ matrices over $\mathbb{F}$, the above shows that S is the inverse of T.

$$T, S \colon \mathbb{F}^n \to \mathbb{F}^n$$

$$TS = I \Leftrightarrow ST = I$$

And this shows that T is invertible with inverse S.

$$I = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}$$

Remark

Upshot

$$(v_1', \ldots, v_n') = (v_1, \ldots, v_n) \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \cdots & S_{nn} \end{pmatrix} \quad \& \quad T = S^{-1}$$

$$(v_1, \ldots, v_n) = (v_1', \ldots, v_n') \begin{pmatrix} T_{11} & \cdots & T_{1n} \\ \vdots & \ddots & \vdots \\ T_{n1} & \cdots & T_{nn} \end{pmatrix}$$

S is the change of basis matrix from

$$v_1, \ldots, v_n \text{ to } v_1', \ldots, v_n'$$

Definition 10.1

Group (for linear algebra)

1. A set with a multiplication (associative).
2. $\exists$ a multiplicative identity.

Remark

Invertible $n \times n$ matrices form a group

Here the multiplication is the product of matrices & the identity is I.
It is called the General linear group $GL(n, \mathbb{F})$, which is the set of all invertible $n \times n$ matrices over $\mathbb{F}$.

10.2.2 Transformation of Matrix Representation

Consider $T : V \to V$

$$\begin{cases} v_1, \ldots, v_n \\ v_1', \ldots, v_n' \end{cases}$$

Are related by a matrix $S \in GL(n, \mathbb{F})$

$$(v_1', \ldots, v_n') = (v_1, \ldots, v_n)S$$

We have the relationship between the coordinates of v_i and v_i'.

$$Tv_i = \sum_{j=1}^{n} v_j T_{ji} \qquad [T_{ij}] \in \mathbb{F}^{n \times n}$$

$$Tv'_i = \sum_{j=1}^{n} v'_j T'_{ji} \qquad [T'_{ij}] \in \mathbb{F}^{n\times n}$$

We can express the action of T on the basis vectors in terms of matrices

$$\begin{aligned} T(v_1,\ldots,v_n) &= (Tv_1,\ldots,Tv_n) = (v_1,\ldots,v_n)\mathcal{M}(T) \\ T(v'_1,\ldots,v'_n) &= (Tv'_1,\ldots,Tv'_n) = (v'_1,\ldots,v'_n)\mathcal{M}'(T) \\ &= (v_1,\ldots,v_n)S\mathcal{M}'(T) \end{aligned}$$

$$\begin{aligned} T(v'_1,\ldots,v'_n) &= T(v_1,\ldots,v_n)S \\ &= (v_1,\ldots,v_n)\mathcal{M}(T)S \end{aligned}$$

So we have

$$(v_1,\ldots,v_n)\mathcal{M}(T)S = (v_1,\ldots,v_n)S\mathcal{M}'(T)$$

So that

$$\mathcal{M}(T)S = S\mathcal{M}'(T)$$

Right multiplication by S^{-1}

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

$$Tv_i = \sum_{j=1}^{n} v_j T_{ji}$$

For the change of basis $v'_i = \sum_{j=1}^{n} v_j S_{ji}$ we have

$$\begin{aligned} Tv'_i &= \sum_{j=1}^{n} v'_j T'_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k S_{kj} \right) T'_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} S_{kj} T'_{ji} \right) \end{aligned}$$

Then we can also express Tv'_i in terms of Tv_j and the change of basis matrix S.

$$\begin{aligned} Tv'_i &= T\left(\sum_{j=1}^{n} v_j S_{ji} \right) \\ &= \sum_{j=1}^{n} (Tv_j) S_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k \mathcal{M}(T)_{kj} \right) S_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} \mathcal{M}(T)_{kj} S_{ji} \right) \end{aligned}$$

So we have

$$\sum_{j=1}^{n} S_{kj}\mathcal{M}'(T)_{ji} = \sum_{j=1}^{n} \mathcal{M}(T)_{kj} S_{ji}$$

$$(S\mathcal{M}'(T))_{ki} = (\mathcal{M}(T)S)_{ki}$$
$$S\mathcal{M}'(T) = \mathcal{M}(T)S$$

Right multiplication by S^{-1} gives

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

10.2.3 Summary

For a linear map T

$$T\colon V \to V \quad (\dim V = n)$$

And two different bases for V

$$\begin{cases} v_1, \dots, v_n \\ v_1', \dots, v_n' \end{cases}$$

We have two different matrix representations of T corresponding to the two bases.

$$\begin{aligned} &\mathcal{M}(T) && \text{rel. to } v_1, \dots, v_n \\ &\mathcal{M}'(T) && \text{rel. to } v_1', \dots, v_n' \end{aligned}$$

Then

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

Where

1. $S \in GL(n, \mathbb{F})$
2. $(v_1', \dots, v_n') = (v_1, \dots, v_n)S$

> **Remark**
>
> Any intrinsic properties of $T\colon V \to V$ must be invariant under change of basis.
>
> - trace of T
> - determinant of T
> - characteristic poly of T
> - $\dots$

10.3 Change of Basis for $T\colon V \to W$

For a linear map

$$T\colon V \to W$$
$$\dim V = n \qquad \dim W = m$$

Choose bases of V and W

$$\left.\begin{matrix} v_1, \dots, v_n \text{ for } V \\ w_1, \dots, w_m \text{ for } W \end{matrix}\right\} \implies \mathcal{M}(T) \in \mathbb{F}^{m,n}$$

$$Tv_i = \sum_{j=1}^{m} w_j T_{ji} \leftarrow \text{ entries of } \mathcal{M}(T)$$

Choose other bases of V and W

$$\left.\begin{array}{ll} v'_1, \ldots, v'_n & \text{for } V \\ w'_1, \ldots, w'_m & \text{for } W \end{array}\right\} \implies \mathcal{M}'(T) \in \mathbb{F}^{m,n}$$

$$Tv'_i = \sum_{j=1}^{m} w'_j \mathcal{M}'(T)_{ji} \leftarrow \text{ entries of } \mathcal{M}'(T)$$

Two change-of-basis matrices

$$\begin{array}{ll} S \in GL(n, \mathbb{F}) & (v'_1, \ldots, v'_n) = (v_1, \ldots, v_n)S \\ R \in GL(m, \mathbb{F}) & (w'_1, \ldots, w'_m) = (w_1, \ldots, w_m)R \end{array}$$

Guess:

$$\underset{m'(T)}{\left(m \times n\right)} = \underset{R^{-1}}{\left(m \times m\right)} \underset{m(T)}{\left(\begin{array}{c}\text{rows} \times \text{columns} \\ m \quad\quad n\end{array}\right)} \underset{S}{\left(n \times n\right)}$$

$$\mathcal{M}'(T) = R^{-1}\mathcal{M}(T)S$$

Then we can verify this by direct computation

$$\begin{aligned} T(v'_1, \ldots, v'_n) = (Tv'_1, \ldots, Tv'_n) &= (w'_1, \ldots, w'_m)\mathcal{M}'(T) \\ &= (w_1, \ldots, w_m)R\mathcal{M}'(T) \end{aligned}$$

$$\begin{aligned} T(v'_1, \ldots, v'_n) &= T(v_1, \ldots, v_n)S \\ &= (w_1, \ldots, w_m)\mathcal{M}(T)S \end{aligned}$$

So we have

$$(w_1, \ldots, w_m)\mathcal{M}(T)S = (w_1, \ldots, w_m)R\mathcal{M}'(T)$$
$$\mathcal{M}(T)S = R\mathcal{M}'(T)$$

Right multiplication by R^{-1} gives

$$\mathcal{M}'(T) = R^{-1}\mathcal{M}(T)S$$

Remark

Last lecture:

$$\exists R \in GL(m, \mathbb{F})$$
$$\exists S \in GL(n, \mathbb{F})$$

$$\mathcal{M}'(T) = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \quad r = \operatorname{rank} T = \dim \operatorname{im} T$$

11 Lecture 11

- Product
- Quotient

Vector spaces are sets of structures

11.1 Product

11.1.1 Definition and Basic Properties

Definition 11.1

V_1 and V_2 are vector spaces over $\mathbb{F}$. The *product* of V_1 and V_2 is the vector space

$$V_1 \times V_2 = \{(v_1, v_2) \mid v_1 \in V_1, v_2 \in V_2\}$$

with addition and scalar multiplication
for $v_1, v_2 \in V_1 \times V_2$ and $\alpha \in \mathbb{F}$

$$(v_1, v_2) + (v_1', v_2') = (v_1 + v_1', v_2 + v_2')$$
$$\alpha(v_1, v_2) = (\alpha v_1, \alpha v_2)$$

Remark

The dimension of the product of two vector spaces is the sum of their dimensions.

$$\dim V_1 = n_1$$
$$\dim V_2 = n_2$$

Claim:

$$\dim(V_1 \times V_2) = n_1 + n_2$$

Proof. Choose

$$\{e_1, \ldots, e_{n_1}\} \text{ is a basis of } V_1$$
$$\{f_1, \ldots, f_{n_2}\} \text{ is a basis of } V_2$$

Then

$$(e_1, 0), \ldots, (e_{n_1}, 0), (0, f_1), \ldots, (0, f_{n_2})$$

is a basis of $V_1 \times V_2$. For $(v_1, v_2) \in V_1 \times V_2$, we can write

$$v_1 = \sum_{i=1}^{n_1} \alpha_i e_i, \quad v_2 = \sum_{j=1}^{n_2} \beta_j f_j, \quad \alpha_i, \beta_j \in \mathbb{F}$$

$$(v_1, v_2) = \sum_{i=1}^{n_1} \alpha_i (e_i, 0) + \sum_{j=1}^{n_2} \beta_j (0, f_j)$$

for some $\alpha_i, \beta_j \in \mathbb{F}$, showing that the set spans $V_1 \times V_2$. Linear independence is straightforward to verify.

$$\text{If } \sum_{i=1}^{n_1} \alpha_i (e_i, 0) + \sum_{j=1}^{n_2} \beta_j (0, f_j) = (0, 0)$$

Then

$$\sum_{i=1}^{n_1} \alpha_i e_i = 0 \text{ and } \sum_{j=1}^{n_2} \beta_j f_j = 0$$

Then all the coefficients α_i and β_j must be zero, proving linear independence. □

Definition 11.2

$V_1, \ldots, V_m$ are vector spaces over $\mathbb{F}$. The *product* of $V_1, \ldots, V_m$ is the vector space over $\mathbb{F}$ defined by

$$V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) \mid v_i \in V_i \text{ for } i = 1, \ldots, m\}$$

The operations on $V_1 \times \cdots \times V_m$ are defined componentwise

$$(v_1, \ldots, v_m) + (v_1', \ldots, v_m') = (v_1 + v_1', \ldots, v_m + v_m')$$

$$\alpha(v_1, \ldots, v_m) = (\alpha v_1, \ldots, \alpha v_m)$$

Theorem 11.1

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$$

Proof.

$$\begin{aligned} &e_1^{(1)}, \ldots, e_{n_1}^{(1)} && \text{is a basis of } V_1 \\ &e_1^{(2)}, \ldots, e_{n_2}^{(2)} && \text{is a basis of } V_2 \\ &\vdots \\ &e_1^{(m)}, \ldots, e_{n_m}^{(m)} && \text{is a basis of } V_m \end{aligned}$$

Then

$$\begin{aligned} &(e_j^{(1)}, 0, \ldots, 0) && j = 1, \ldots, n_1 \\ &(0, e_j^{(2)}, \ldots, 0) && j = 1, \ldots, n_2 \\ &\vdots \\ &(0, \ldots, 0, e_j^{(m)}) && j = 1, \ldots, n_m \end{aligned}$$

is a basis of $V_1 \times \cdots \times V_m$, proving that

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$$

□

11.1.2 Relation to Direct Sum

Relate the product to the direct sum

Theorem 11.2

Let $V_1, \ldots, V_m$ are subspaces of a vector space V. We can define the linear map

$$\begin{aligned} \phi : V_1 \times \cdots \times V_m &\to V \\ (v_1, \ldots, v_m) &\mapsto v_1 + \cdots + v_m \end{aligned}$$

The sum $V_1 + \cdots + V_m$ is direct $\iff$ ϕ is an isomorphism.

Remark

Direct

$$v_1 + \cdots + v_m = 0 \iff v_1 = \cdots = v_m = 0$$

for all $v_i \in V_i$.

Proof. Proof needed for both directions of the equivalence.

($\Longrightarrow$) Suppose $\phi(v_1, \ldots, v_m) = 0$

1. ϕ is injective
   (by definition of direct sum)
2. ϕ is surjective
   (by construction)

$\Longrightarrow$ ϕ is an isomorphism.

($\Longleftarrow$) Suppose ϕ is an isomorphism. Then ϕ is injective

$$\phi(v_1, \ldots, v_m) = 0 \Longrightarrow (v_1, \ldots, v_m) = 0$$

Hence

$$v_1 + \cdots + v_m = 0 \Longrightarrow v_1 = \cdots = v_m = 0$$

for all $v_i \in V_i$, proving that the sum is direct. □

Corollary 11.3

$$\dim(V_1 \oplus \cdots \oplus V_m) = \sum_{i=1}^{m} \dim V_i$$

11.2 Quotient

11.2.1 Equivalence Relation

Definition 11.3

Define an equivalence relation on V:

$$v_1 \sim v_2 \iff v_1 - v_2 \in U.$$

an equivalence relation satisfying reflexivity, symmetry, and transitivity.

1. $v \sim v$
$$v - v = 0 \in U$$
2. If $v_1 \sim v_2$, then $v_2 \sim v_1$
$$v_2 - v_1 = -(v_1 - v_2) \in U v_1 - v_2 \in U$$
3. If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 \sim v_3$
$$v_1 - v_2 \in U v_2 - v_3 \in U$$
So
$$v_1 - v_3 = (v_1 - v_2) + (v_2 - v_3) \in U$$
$$v_1 - v_3 \in U$$

Definition 11.4

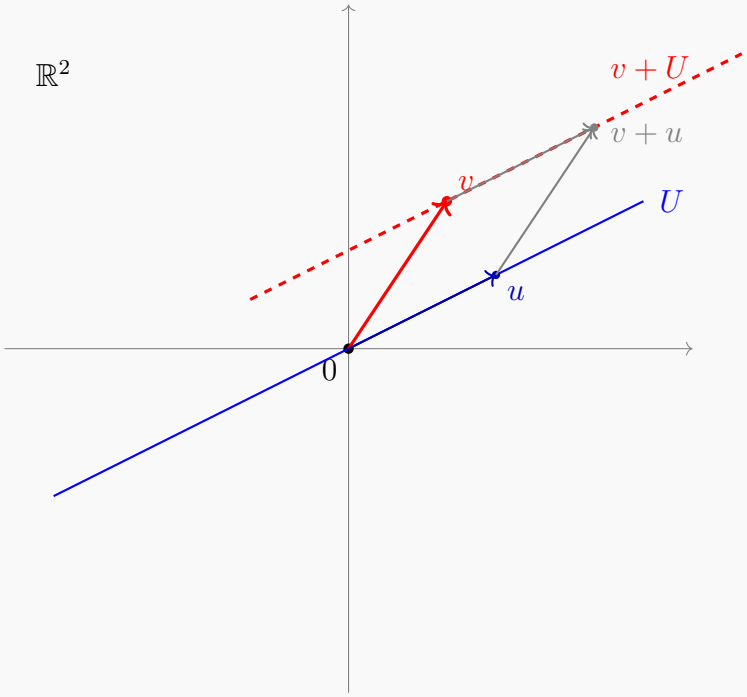
The equivalence class of a vector $v \in V$ under the relation $\sim$ is the set

$$[v] = \{w \in V \mid w \sim v\} = v + U.$$

is the translate of the subspace U by the vector v.

Example

Let $V = \mathbb{R}^2$ and U be a subspace.



11.2.2 The quotient

Definition 11.5

The quotient space V/U is the set of equivalence classes.

Let V be a vector space and U a subspace of V. The quotient space V/U is the set of cosets of U in V:

$$V/U = \{v + U \mid v \in V\}.$$

Definition 11.6

The quotient map

The quotient map is the map

$$\begin{aligned} \pi \colon V &\to V/U \\ v &\mapsto [v] = v + U \end{aligned}$$

Theorem 11.4

Claim V/U can be made into a vector space such that π is linear

$$[v] = v + U$$
$$[w] = w + U$$

For $\lambda \in \mathbb{F}$

$$[v] + [w] = [v + w]$$
$$\lambda[v] = [\lambda v]$$

Proof. We need to show two things to verify that this definition of addition and scalar multiplication on V/U are proper.

1. *Operation well defined*

   First, we need to check that the operations are well defined

   $$\begin{aligned} [v] = [v'] &\implies v - v' \in U \\ &\implies \exists u \in U \text{ s.t. } v - v' = u \\ &\implies v = v' + u \end{aligned}$$

   So

   $$[\lambda v] = [\lambda(v' + u)] = [\lambda v' + \lambda u] = [\lambda v']$$

   We have

   $$\lambda v \sim \lambda v' \implies [\lambda v] = [\lambda v']$$

   Similarly, addition is well defined.

   $$\begin{aligned} [v] = [v'] &\implies v - v' \in U \\ [w] = [w'] &\implies w - w' \in U \\ &\implies (v + w) - (v' + w') = (v - v') + (w - w') \in U \\ &\implies [v + w] = [v' + w'] \end{aligned}$$

2. *π is linear*

   Then we check

   $$\pi \colon V \to V/U$$
   $$v \mapsto [v]$$

   is linear Let $v, w \in V$ and $\lambda \in \mathbb{F}$. Then

   $$\pi(v + w) = [v + w] = [v] + [w] = \pi(v) + \pi(w),$$

   and

   $$\pi(\lambda v) = [\lambda v] = \lambda[v] = \lambda\pi(v).$$

   Hence, π is linear.

□

Remark

well-defined is the operation being independent of the representative chosen.

Lemma 11.5

The canonical projection $\pi\colon V \to V/U$ is surjective.

Proof. Let

$$[v] \in V/U$$

By definition of the quotient space

$$[v] = v + U$$

for some $v \in V$. Then

$$\pi(v) = [v]$$

showing that every element of V/U is in the image of π.

$$\operatorname{im}(\pi) = V/U$$

□

Lemma 11.6

The kernel of the canonical projection $\pi\colon V \to V/U$ is U.

Proof. Let $v \in V$. Then

$$\begin{aligned} v \in \ker(\pi) &= \{\ v \in V \mid \pi(v) = [0]\ \} \\ &= \{\ v \in V \mid [v] = [0]\ \} \\ &= U \end{aligned}$$

So that we have

$$v - 0 \in U \iff v \in U$$

Hence

$$\ker(\pi) = U$$

□

Theorem 11.7

V is finite dimensional

$$\dim V/U = \dim V - \dim U$$

Proof. Use rank theorem since π is linear, we have

$$\dim V = \dim(\ker(\pi)) + \dim(\operatorname{im}(\pi))$$

And

$$\begin{aligned} \ker(\pi) &= U \\ \operatorname{im}(\pi) &= V/U \end{aligned}$$

So we have

$$\dim V = \dim U + \dim V/U \implies \dim V/U = \dim V - \dim U$$

□

12 Lecture 12

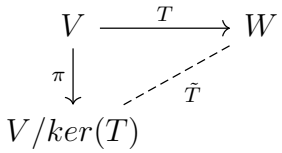
Dual spaces and annihilators

12.1 Last Lecture

For $U \subseteq V$

$$\begin{aligned} V &\xrightarrow{\pi} V/U \\ v &\mapsto [v] = v + U \end{aligned}$$

We have the following commutative diagram



$U = \ker(T) \subseteq V$ The next we have the definition of the induced map

Definition 12.1

$$\tilde{T} \colon V/\ker(T) \to W$$

defined by

$$\tilde{T}([v]) = T(v)$$

for all $v \in V$, is called the *induced map* or *quotient map* of T by $\ker(T)$.

Lemma 12.1

The kernel and image of the induced map $\tilde{T}$ are given by

$$\begin{cases} \ker(\tilde{T}) &= \{[0]\} \\ \operatorname{im}(\tilde{T}) &= \operatorname{im}(T) \end{cases}$$

Proof. For the kernel, we have

$$\begin{aligned} \ker(\tilde{T}) &= \{[v] \in V/\ker(T) \mid \tilde{T}([v]) = 0\} \\ &= \{[v] \in V/\ker(T) \mid T(v) = 0\} \\ &= \ker(T)/\ker(T) \\ &= \{[0]\} \end{aligned}$$

For the image, we have

$$\begin{aligned} \operatorname{im}(\tilde{T}) &= \{\tilde{T}([v]) \mid [v] \in V/\ker(T)\} \\ &= \{T(v) \mid v \in V\} \\ &= \operatorname{im}(T) \end{aligned}$$

□

12.2 Dual Spaces

12.2.1 Dual Space and Dual Basis

Let V and W be finite dimensional vector spaces over a field $\mathbb{F}$.

V, W	$\mathbb{F}$	finite dimensional
$\mathcal{L}(V, W)$	$\mathbb{F}$	$\dim \mathcal{L}(V, W) = \dim V \dim W$
V'	$\mathbb{F}$	$\dim V' = \dim V$

For the spaces above, when $W = \mathbb{F}$, we write $V' = \mathcal{L}(V, \mathbb{F})$ for the dual space of V.

Definition 12.2

The dual space of a vector space V over a field $\mathbb{F}$ is the vector space $V' = \mathcal{L}(V, \mathbb{F})$ consisting of all linear maps from V to $\mathbb{F}$.

Definition 12.3

Let $v_1, \ldots, v_n$ be a basis for V. The *(Canonical) dual basis* of V' corresponding to the basis $v_1, \ldots, v_n$ is $\varphi_1, \ldots, \varphi_n \in V'$ defined by

$$\varphi_i(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

So, for any vector $v \in V$, we can express it in terms of the basis $v_1, \ldots, v_n$ as

$$v = a_1 v_1 + \cdots + a_n v_n, \quad a_i \in \mathbb{F}$$

$$\varphi_i(v) = a_1 \varphi_i(v_1) + \cdots + a_n \varphi_i(v_n) = a_i$$

$$v = \varphi_1(v) v_1 + \cdots + \varphi_n(v) v_n$$

Theorem 12.2

The canonical dual basis

$$\varphi_1, \ldots, \varphi_n$$

is a basis for V'.

Proof.

$$\dim V' = \dim V = n \quad \& \quad \varphi_1, \ldots, \varphi_n$$

This follows from linear independence.

$$a_1 \varphi_1 + \cdots + a_n \varphi_n = 0$$

$$\iff \forall v \in V \ (a_1 \varphi_1 + \cdots + a_n \varphi_n)(v) = 0$$

Then $\forall i$

$$0 = (a_1 \varphi_1 + \cdots + a_n \varphi_n)(v_i) = a_i$$

□

12.2.2 Dual map

For a linear map $T : V \to W$ and a linear functional $\varphi \in W'$, the composition $\varphi \circ T$ is a linear functional in V'. We have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow^{\varphi\circ T} & \downarrow{\scriptstyle \varphi} \\ & & \mathbb{F} \end{array}$$

So that we can define the dual map from W' to V'

Definition 12.4

For a linear map $T : V \to W$, the dual map $T' \colon W' \to V'$ is defined by

$$\begin{aligned} T' \colon W' &\to V' \\ \varphi &\mapsto T'(\varphi) = \varphi \circ T \end{aligned}$$

Which is

$$W' \xrightarrow{T'} V'$$

Lemma 12.3

The dual map T' is linear.

Proof. $\forall \varphi, \psi \in W', \forall \lambda \in \mathbb{F}$

$$\begin{aligned} T'(\varphi + \lambda\psi)(v) &= (\varphi + \lambda\psi)(Tv) \\ &= \varphi(Tv) + \lambda\psi(Tv) \\ &= (T'\varphi)(v) + \lambda(T'\psi)(v) \\ &= (T'\varphi + \lambda T'\psi)(v). \end{aligned}$$

Hence

$$T'(\varphi + \lambda\psi) = T'\varphi + \lambda T'\psi$$

linearity □

Example

Consider the differentiation operator and evaluation at a point $x_0 \in \mathbb{R}$ which is a linear functional.

$$\begin{aligned} \mathcal{D} \colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto p'(x) \end{aligned}$$

$$\begin{aligned} \varphi \colon \mathcal{P}(\mathbb{R}) &\to \mathbb{R} \\ p(x) &\mapsto p(x_0) \end{aligned}$$

We have

$$(\mathcal{D}\varphi)(p) = \varphi(\mathcal{D}p) = \varphi(p') = p'(x_0)$$

Define $\varphi(p) = \int_0^1 p(x)dx$

$$(\mathcal{D}\varphi)(p) = \varphi(p') = \int_0^1 p'(x)dx = p(1) - p(0)$$

12.2.3 Linearity of Dual Operation

For linear maps T

$$V \xrightarrow{T} W \qquad W' \xrightarrow{T'} V'$$

Their action on dual maps is given by

$$\begin{aligned} \mathcal{L}(V,W) &\to \mathcal{L}(W',V') \\ T &\mapsto T' \end{aligned}$$

Which is a linear map from $\mathcal{L}(V,W)$ to $\mathcal{L}(W',V')$.

Definition 12.5

The dual operation $T \mapsto T'$ is defined for linear maps $T \in \mathcal{L}(V,W)$ as the map $T' \in \mathcal{L}(W',V')$ given by

$$(T'\varphi)(v) = \varphi(Tv)$$

Lemma 12.4

The dual operation

$$T \mapsto T'$$

is linear.

Proof. For $S, T \in \mathcal{L}(V,W)$ and $\lambda \in \mathbb{F}$, Check

$$(S+\lambda T)' = S' + \lambda T'$$

For any $v \in V$ and $\varphi \in W'$,

$$\begin{aligned} ((S+\lambda T)'\varphi)(v) &= \varphi((S+\lambda T)(v)) \\ &= \varphi(Sv + \lambda Tv) \\ &= \varphi(Sv) + \lambda\varphi(Tv) \\ &= (S'\varphi + \lambda T'\varphi)(v) \end{aligned}$$

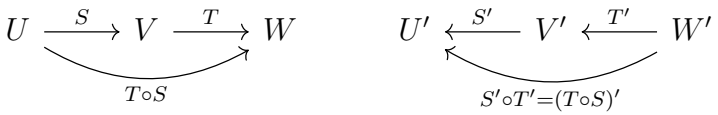
So that

$$(S+\lambda T)' = S' + \lambda T'$$

□

12.2.4 Composition of Dual Maps

For linear maps $S\colon U \to V$ and $T\colon V \to W$, we have following diagrams



Theorem 12.5

For the composition of dual maps, we have

$$(T \circ S)' = S' \circ T'$$

Proof. For any $\varphi \in W'$

$$\begin{aligned}(S' \circ T')(\varphi) = S'(T'\varphi) &= (T'\varphi) \circ S \\ &= (\varphi \circ T) \circ S \\ &= \varphi \circ (T \circ S) \\ &= (T \circ S)'(\varphi)\end{aligned}$$

□

Notation

The composition of dual maps reverses the order of composition

$$(T \circ S)' = S' \circ T'$$

Has such similarity to the transposition of matrices

$$(AB)^T = B^T A^T$$

And in fact, through choosing appropriate bases, the dual map corresponds to the transpose of the matrix representing the original linear map.

$$T \colon V \to W \qquad \begin{cases} v_1, \dots, v_n \\ w_1, \dots, w_m \end{cases} \qquad \mathcal{M}(T)$$

$$T' \colon W' \to V' \qquad \begin{cases} \psi_1, \dots, \psi_m \\ \varphi_1, \dots, \varphi_n \end{cases} \qquad \mathcal{M}(T') = \mathcal{M}(T)^t$$

12.3 Annihilators

Definition 12.6

Let V be a vector space over a field $\mathbb{F}$ and

$$U \subseteq V$$

The *annihilator* of U in V' is defined as

$$U^0 = \{\varphi \in V' \mid \varphi(u) = 0 \ \forall u \in U\}$$

Example

$V = \mathbb{R}^3$ and let

$$U = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$$

Then $U^0 = ?$

Let $\varphi_1, \varphi_2, \varphi_3$ and e_1, e_2, e_3 be the canonical dual basis of $(\mathbb{R}^3)'$ and basis of $\mathbb{R}^3$.

$$(x_1, x_2, x_3) = x_1 e_1 + x_2 e_2 + x_3 e_3$$

Let $\varphi \in (\mathbb{R}^3)'$

$$\varphi = a_1\varphi_1 + a_2\varphi_2 + a_3\varphi_3$$

Then

$$\varphi \in U^0 \Leftrightarrow \varphi(x_1, x_2, -x_1 - x_2) = 0$$

$$a_1 x_1 + a_2 x_2 + a_3(-x_1 - x_2) = 0$$

$$(a_1 - a_3)x_1 + (a_2 - a_3)x_2 = 0$$

So we must have

$$a_1 - a_3 = 0$$

$$a_2 - a_3 = 0$$

$$a_1 = a_2 = a_3$$

$$\varphi = a_1(\varphi_1 + \varphi_2 + \varphi_3)$$

Therefore

$$U^0 = \{a(\varphi_1 + \varphi_2 + \varphi_3) \mid a \in \mathbb{R}\}$$

Theorem 12.6

The annihilator which is

$$U^0 \subseteq V'$$

is a subspace of V'

Proof. Let $\varphi, \psi \in U^0$ and $\lambda \in \mathbb{F}$

$$\varphi + \lambda\psi \overset{?}{\in} U^0$$

Let $u \in U$

$$(\varphi + \lambda\psi)(u) = \overset{=0}{\varphi(u)} + \lambda\overset{=0}{\psi(u)} = 0$$

So indeed

$$\varphi + \lambda\psi \in U^0$$

□

Theorem 12.7

For subspace $U \subseteq V$ of a finite-dimensional vector space V $(\dim V < \infty)$

$$\dim U^0 = \dim V - \dim U$$

Remark

Notice that

$$\dim U^0 = \dim V - \dim U = \dim V/U.$$

Proof. Start with basis of U

$$u_1, \dots, u_n \ \in U$$

Extend this to a basis of V

$$u_1, \dots, u_n, v_1, \dots, v_m \ \in V$$

Consider the dual basis of V'

$$\varphi_1, \dots, \varphi_n, \psi_1, \dots, \psi_m \ \in V'$$

We claim that

$$U^0 = \operatorname{span}\{\psi_1, \dots, \psi_m\}$$

So that

$$\begin{aligned} \dim U^0 &= m \\ &= m + n - n \\ &= \dim V - \dim U \end{aligned}$$

Then we need to prove the claim that $\psi_1, \dots, \psi_m$ is a basis of U^0.

NB $\psi_1, \dots, \psi_m$ are linearly independent by definition

$$\forall u \in U \quad u = a_1 u_1 + \cdots + a_n u_n$$

Apply ψ_i to u gives

$$\psi_i(u) = a_1 \psi_i(u_1) + \cdots + a_n \psi_i(u_n) = 0 \implies \psi_i \in U^0$$

For any $\varphi \in U^0 \subseteq V'$

$$\varphi = a_1 \varphi_1 + \cdots + a_n \varphi_n + \underbrace{b_1 \psi_1 + \cdots + b_m \psi_m}_{\in U^0}$$

So

$$0 = \varphi(u_i) = a_i \quad \forall i = 1, \dots, n$$

And hence

$$\begin{aligned} \varphi = b_1 \psi_1 + \cdots + b_m \psi_m &\implies \psi_1, \dots, \psi_m \text{ span } U^0 \\ &\implies \psi_1, \dots, \psi_m \text{ is a basis of } U^0 \end{aligned}$$

□

Corollary 12.8

For a subspace U of a finite-dimensional vector space V, we have the following facts

$$\begin{aligned} U^0 = \{0\} &\iff U = V \\ U^0 = V' &\iff U = \{0\} \end{aligned}$$

Theorem 12.9

For finite-dimensional vector spaces V and W

$$V \xrightarrow{T} W \quad W' \xrightarrow{T'} V'$$

1. $\ker T' = (\operatorname{im} T)^0$
2. $\operatorname{im} T' = (\ker T)^0$
3. $\dim \operatorname{im} T' = \dim \operatorname{im} T$
4. $\dim \operatorname{im} T' = \dim \ker T + \dim W - \dim V$

13 Lecture 13

Week 8, Lecture 22, Eigenvalues and Eigenvectors

13.1 Invariant Subspaces

Definition 13.1

A linear operator T on V is a linear transformation,

$$T \in \mathcal{L}(V) = \mathcal{L}(V, V)$$

Definition 13.2

Let U be a subspace of V, $U \subseteq V$. U is called *invariant* under the linear operator $T \in \mathcal{L}(V)$ if

$$T(U) \subseteq U.$$

Example

For $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$,

$$T(p(x)) = p'(x)$$

$\mathcal{P}_4(\mathbb{R})$ is invariant under T.

$$\deg(T(p)) = \deg(p') \leq \deg(p) \leq 4$$

Example

$$\text{null } T \text{ is invariant under } T.$$
$$\text{Range } T \text{ is invariant under } T.$$

13.2 Eigenvalue and Eigenvector

Definition 13.3

Suppose $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is called an *eigenvalue* of T if there exists a nonzero vector $v \in V$ such that

$$T(v) = \lambda v.$$

The vector v is called an *eigenvector* of T corresponding to the eigenvalue λ.

Theorem 13.1

Equivalent conditions for eigenvalues

1. λ is an eigenvalue of T.
2. $T - \lambda I$ is not injective.
3. $T - \lambda I$ is not surjective.
4. $T - \lambda I$ not invertible.

Proof.

$$\begin{aligned} T(v) = \lambda v &\iff (T - \lambda I)(v) = 0 \text{ for some } v \neq 0 \\ &\iff T - \lambda I \text{ is not injective} \\ &\iff T - \lambda I \text{ is not surjective} \\ &\iff T - \lambda I \text{ is not invertible.} \end{aligned}$$

It implies

$$v \neq 0 \in \operatorname{null}(T - \lambda I)$$

So

$$\operatorname{null}(T - \lambda I) \neq \{0\}.$$

The $T - \lambda I$ is not injective, since the rank theorem implies

$$\operatorname{range}(T - \lambda I) + \operatorname{null}(T - \lambda I) = \dim V,$$

and

$$\dim \operatorname{null}(T - \lambda I) \neq 0.$$

Means the operator $T - \lambda I$ is not surjective, therefore, the operator $T - \lambda I$ is not invertible. □

Example

For the linear transformation $T \in \mathcal{L}(\mathbb{F}^2)$ defined by

$$T(w, z) = (-z, w).$$

The matrix representation of T with respect to the standard basis is

$$[T] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

continued

The geometric interpretation of T is that it rotates the vector (w, z) by swapping its components.

- if $\mathbb{F} = \mathbb{R}$, no vector is left in the same direction by T, so no

$$T(v) = \lambda v.$$

  So there are no real eigenvalues of T.

- if $\mathbb{F} = \mathbb{C}$, the eigenvalues of T are i and $-i$.

$$\begin{aligned} T(w, z) = \lambda(w, z) &\implies (z, w) = \lambda(w, z) \\ &\implies -z = \lambda w, w = \lambda z \\ &\implies \lambda^2 = -1 \\ &\implies \lambda = \pm i. \end{aligned}$$

Eigenvectors corresponding to the eigenvalues i and $-i$ can be found by solving (for $\lambda = i$ and $\lambda = -i$) the equation

$$(-z, w) = \lambda(w, z)$$

- For $\lambda = i$, we have

$$\begin{aligned} (-z, w) = i(w, z) &\implies -z = iw, w = iz \\ &\implies w = iz, z = -iw, \end{aligned}$$

  which gives the eigenvector $(w, z) = (1, -i)$ up to scalar multiples.

- For $\lambda = -i$, we have

$$\begin{aligned} (-z, w) = -i(w, z) &\implies -z = -iw, w = -iz \\ &\implies w = -iz, z = iw, \end{aligned}$$

  which gives the eigenvector $(w, z) = (1, i)$ up to scalar multiples.

13.3 Observations on Eigenvalues and Eigenvectors

Theorem 13.2

Eigenvectors corresponding to distinct eigenvalues of a linear operator T are linearly independent.

Proof. Let

$\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of T,

$v_1, v_2, \ldots, v_n$ are the corresponding eigenvectors of T.

If

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0.$$

Apply $T - \lambda_m I$ to the equation,

$$a_1(T - \lambda_m I)v_1 + a_2(T - \lambda_m I)v_2 + \cdots + a_n(T - \lambda_m I)v_n = 0,$$
$$a_1(\lambda_1 - \lambda_m)v_1 + a_2(\lambda_2 - \lambda_m)v_2 + \cdots + a_n(\lambda_n - \lambda_m)v_n = 0.$$

Suppose we have linearly dependent list of eigenvectors, say

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0.$$

With some

$$a_i \neq 0.$$

Then applying $T - \lambda_m I$ as above leads to a contradiction because the coefficient of v_m would be nonzero. Therefore, the eigenvectors corresponding to distinct eigenvalues are linearly independent. □

Theorem 13.3

Consider a vector space V over $\mathbb{F}$ with

$$\dim V = n$$

let $T \in \mathcal{L}(V)$ be a linear operator.
Then T has at most n distinct eigenvalues.

13.4 Polynomial and Operator

Definition 13.4

Let $T \in \mathcal{L}(V)$ and

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathcal{P}(\mathbb{F}).$$

The polynomial p applied to the operator T is defined as

$$p(T) = a_0 I + a_1 T + \cdots + a_n T^n,$$

where I is the identity operator on V.

Property

Let $p(x), q(x) \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then

$$(p + q)(T) = p(T) + q(T),$$

$$1(T) = I,$$

where 1 denotes the constant polynomial 1.

Theorem 13.4

Let $T \in \mathcal{L}(V)$ and $p(x), q(x) \in \mathcal{P}(\mathbb{F})$. Then

$$(pq)(T) = p(T)q(T),$$

and hence

$$p(T)q(T) = q(T)p(T).$$

Corollary 13.5

Let $T \in \mathcal{L}(V)$ and $p(x), q(x) \in \mathcal{P}(\mathbb{F})$, then for $k \in \mathbb{Z}^+$, we have

$$p^0(T) = I,$$

$$(p^k)(T) = \big(p(T)\big)^k,$$

and

$$\big((p+q)^k\big)(T) = \big(p(T) + q(T)\big)^k.$$

Theorem 13.6

Suppose $T \in \mathcal{L}(V)$ and $p(x) \in \mathcal{P}(\mathbb{F})$.

$$\ker(p(T)) \text{ and } \operatorname{im}(p(T)) \text{ are invariant under T .}$$

Proof.

- Let $v \in \text{null}(p(T))$, then

  $$p(T)v = 0.$$

  Applying T to both sides gives

  $$T(p(T)v) = T0 = 0.$$

  Since T and $p(T)$ commute, we have

  $$p(T)(Tv) = 0,$$

  $$Tv \in \text{null}(p(T)).$$

  Hence, the null space of $p(T)$ is invariant under T.

- Let $w \in \text{range}(p(T))$, then there exists $v \in V$ such that

  $$w = p(T)v.$$

  Applying T gives

  $$Tw = T(p(T)v) = p(T)(Tv) \in \text{range}(p(T)).$$

  Hence, the range space of $p(T)$ is invariant under T.

□

14 Lecture 14

Lecture 23, Existence of Eigenvalues

14.1 Existence of Eigenvalues

Theorem 14.1

Every operator on a finite dimensional complex vector space has at least one eigenvalue.

Proof. V is a finite dimensional complex vector space of diminsion

$$\dim V = n > 0.$$

Consider the sequence of vectors

$$v, Tv, T^2v, \ldots, T^nv.$$

Since $\dim V = n$, these $n+1$ vectors are linearly dependent. So there exists a (non-constant) polynomial p (smallest degree) such that

$$p(T)v = 0.$$

By F.T.A there exists a $\lambda \in \mathbb{C}$ such that

$$p(\lambda) = 0.$$

Then there exists a polynomial $q(z) \in \mathcal{P}(\mathbb{C})$ such that

$$p(z) = (z - \lambda)q(z).$$

Hence

$$p(T)v = (T - \lambda I)q(T)v = 0.$$

Because $p(z)$ is the smallest degree polynomial such that $p(T)v = 0$, and

$$\deg q(z) < \deg p(z).$$

So we must have

$$q(T)v \neq 0.$$

Let $w = q(T)v$, then

$$(T - \lambda I)w = 0 \implies Tw = \lambda w.$$

Hence, λ is an eigenvalue of T. □

Remark

The hyposises

1. V is a finite dimensional complex vector space.
2. V is a complex vector space.
3. Smallest degree polynomial $p(z)$.

- The first hypo make the list

$$v, Tv, T^2v, \ldots, T^nv.$$

  is linearly dependent since it contains $n+1$ vectors in an n-dimensional space.

- The second hypo is nessary to apply the Fundamental Theorem of Algebra to the polynomial $p(z)$.

- The third hypo ensures that

$$q(T)v \neq 0,$$

  which is necessary to construct the eigenvector w.

14.2 Minimal Polynomial

Theorem 14.2

Suppose V is finite dimensional and $T \in \mathcal{L}(V)$. There is a unique monic polynomial $p(z)$ of minimal degree such that

$$p(T) = 0.$$

Moreover,

$$\deg p \leq \dim V,$$

and p is called the *minimal polynomial* of T.

Proof.

- (Existence) By induction on $\dim V$, we can construct a non-zero polynomial $p(z)$ such that

$$p(T) = 0.$$

- (Uniqueness) Take $p - r$ of some degree (monic)

$$\deg(p - r) < \deg(p)$$

□

Definition 14.1

Minimal Polynomial

Suppose V is finite dimensional and $T \in \mathcal{L}(V)$. The minimal polynomial of T is the unique monic polynomial $p(z)$ of minimal degree such that

$$p(T) = 0.$$

Theorem 14.3

Suppose V is finite dimensional and $T \in \mathcal{L}(V)$.

1. The zeros of the minimal polynomial of T are precisely the eigenvalues of T.
2. If V is complex and the distinct eigenvalues of T are $\lambda_1, \ldots, \lambda_m$, then the minimal polynomial has the form

$$p(z) = \prod_{i=1}^{m} (z - \lambda_i)^{m_i}, \quad m_i \geq 1.$$

In particular, p is a product of distinct linear factors if and only if T is diagonalizable.

Proof. Let p be the minimum polynomial of T.

1. Let $\lambda \in \mathbb{F}$, $p(\lambda) = 0$,

$$p(z) = (z - \lambda)q(z) \text{ where } q \text{ is monic}$$

Since $p(T) = 0$, we have

$$p(T) = (T - \lambda I)q(T) = 0 \text{ for all } v \in V.$$

Since $\deg(q) < \deg(p)$,

$$q(T)v \neq 0$$

So

$$(T - \lambda I)(q(T)v) = 0 \implies \lambda \text{ is an eigenvalue of } T.$$

Let $\lambda \in \mathbb{F}$ be an eigenvalue of T,

$$\exists v \in V \setminus \{0\} \text{ such that } Tv = \lambda v.$$

Then by induction on k, we have

$$T^k v = \lambda^k v \quad \forall k \in \mathbb{N}$$

Now let $p(z) = \sum_{k=0}^{n} a_k z^k$. Then

$$p(T)v = p(\lambda)v$$

p is the minimal polynomial of T, so

$$p(T) = 0.$$

Since $v \neq 0$, we must have

$$p(\lambda) = 0.$$

2. Fundamental theorem of akgebra on $\mathbb{C}$ directly implies this statement, since every non-constant polynomial over $\mathbb{C}$ has a root.

□

14.3 Companion Matrix and Minimal Polynomial

Let $a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}$.
Define

$$M = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}$$

Called the *companion matrix* of the polynomial $z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$.

- 1 appears on the line below the diagonal.
- $-a_k$ appears in the last column of the k-th row.

e_k standard basis

$$Te_1 = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = e_2$$

In general, for $1 \leq k \leq n-1$,

$$Te_k = e_{k+1},$$
$$T^k e_1 = e_{k+1}.$$

And for $k = n$,

$$Te_n = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} -a_0 \\ -a_1 \\ \vdots \\ -a_{n-1} \end{pmatrix} \text{(Last column of the matrix)}$$

Remark

Our goal is find a smallest m s.t.

$$C_0 I + C_1 T + \cdots + C_{m-1}T^{m-1} + T^m = 0.$$

From above

$$\begin{aligned} T^n e_1 &= \begin{pmatrix} -a_0 \\ -a_1 \\ \vdots \\ -a_{n-1} \end{pmatrix} \\ &= -a_0 e_1 - a_1 e_2 - \cdots - a_{n-1} e_n \\ &= (-a_0 I - a_1 T - \cdots - a_{n-1} T^{n-1}) e_1 \end{aligned}$$

Let

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0,$$

then

$$p(T) = T^n + a_{n-1}T^{n-1} + \cdots + a_1 T + a_0 I,$$
$$p(T)e_1 = 0.$$

$p(T)$ is the polynomial in T that annihilates e_1, we have

$$p(T)T = Tp(T)$$

$$\boxed{p(T) \text{ commutes with } T}$$

Then for any $k = 0, 1, \ldots, n-1$,

$$\begin{aligned} p(T)e_{k+1} &= p(T) \cdot T^k e_1 \\ &= T^k p(T)e_1 \\ &= 0. \end{aligned}$$

Because $e_1, e_2, \ldots, e_n$ form a basis of the vector space V, we have

$$p(T) = 0.$$

It remains to prove that no smaller m works. Suppose there is a monic polynomial

$$q(z) = z^d + b_{d-1}z^{d-1} + \cdots + b_1 z + b_0, \qquad d < n,$$

such that

$$q(T) = 0.$$

Then

$$\begin{aligned} 0 = q(T)e_1 &= T^d e_1 + b_{d-1}T^{d-1}e_1 + \cdots + b_1 T e_1 + b_0 e_1 \\ &= e_{d+1} + b_{d-1}e_d + \cdots + b_1 e_2 + b_0 e_1. \end{aligned}$$

But e_{d+1} has coefficient 1 on the right-hand side, while all the other terms lie in

$$\text{span}\{e_1, \ldots, e_d\}.$$

This contradicts the linear independence of $e_1, \ldots, e_{d+1}$. Hence no monic annihilating polynomial of degree $< n$ exists. Therefore the smallest m is

$$\boxed{m = n}.$$

15 Lecture 15

16 Lecture 16

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workshop

20 Lecture 20

Self-adjoint and normal operators

Notation

$$\mathbb{F} = \mathbb{R} \text{ or } \mathbb{C}$$

U, V, W is Inner product space over $\mathbb{F}$ with inner product $\langle \cdot, \cdot \rangle$

20.1 Adjoint

Definition 20.1

The *Adjoint* of a linear transformation $T\colon V \to W$ is the unique linear transformation $T^*\colon W \to V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle \quad \forall v \in V, w \in W.$$

Theorem 20.1

Riesz representation theorem

Let $V \xrightarrow{\varphi} \mathbb{F}$ be a bounded linear functional. Then there exists a unique $v \in V$ such that

$$\varphi(w) = \langle w, v \rangle \quad \forall w \in V.$$

Theorem 20.2

Consider $T \in \mathcal{L}(V, W)$. The adjoint

$$T^* \in \mathcal{L}(W, V)$$

exists and is unique.

Then we use the Riesz representation theorem to prove the existence and uniqueness of the adjoint T^*.

Proof. Take $w \in W$ and consider the linear functional

$$\begin{aligned} V &\to \mathbb{F} \\ v &\mapsto \langle Tv, w \rangle. \end{aligned}$$

By the Riesz representation theorem (RRT), there exists a unique $T^* w \in V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle \quad \forall v \in V.$$

Therefore, there exists a unique $T^* w \in V$ satisfying the adjoint relation. □

Theorem 20.3

If $T \in \mathcal{L}(V, W)$, then the adjoint

$$T^* \in \mathcal{L}(W, V)$$

is linear.

Proof. Let $w_1, w_2 \in W$ and $\lambda \in \mathbb{F}$. Consider $T^*(w_1 + \lambda w_2)$. For all $v \in V$, the adjoint definition gives

$$\langle v, T^*(w_1 + \lambda w_2)\rangle = \langle Tv, w_1 + \lambda w_2\rangle.$$

By the conjugate-linearity of the inner product in the second argument, we get

$$\langle Tv, w_1 + \lambda w_2\rangle = \langle Tv, w_1\rangle + \overline{\lambda}\langle Tv, w_2\rangle.$$

Using the definition of the adjoint again yields

$$\langle Tv, w_1\rangle + \overline{\lambda}\langle Tv, w_2\rangle = \langle v, T^*w_1\rangle + \overline{\lambda}\langle v, T^*w_2\rangle = \langle v, T^*w_1 + \overline{\lambda}T^*w_2\rangle.$$

Since this holds for all $v \in V$, the non-degeneracy of the inner product implies

$$T^*(w_1 + \lambda w_2) = T^*w_1 + \overline{\lambda}T^*w_2.$$

□

Property

$$S, T \colon V \to W, \quad \lambda \in \mathbb{F}, \quad R \colon W \to U$$

1. $(S + \lambda T)^* = S^* + \overline{\lambda}T^*$
2. $(T^*)^* = T$
3. $(R \circ T)^* = T^* \circ R^*$
4. $I_V^* = I_V$
5. if T is invertible, then $(T^*)^{-1} = (T^{-1})^*$

Proof. (We prove (5) here) Let T be invertible, then

$$T^{-1}T = I_V.$$

Taking the adjoint of both sides,

$$(T^{-1}T)^* = (I_V)^*.$$

Using the properties of the adjoint,

$$T^*(T^{-1})^* = I_V.$$

Therefore, $(T^{-1})^*$ is the inverse of T^*, i.e.,

$$(T^*)^{-1} = (T^{-1})^*.$$

□

Definition 20.2

For Inner Product Vector Space V and $U \subseteq V$.

The orthogonal complement of U in V is defined as

$$U^{\perp} = \{v \in V \mid \langle v, u \rangle = 0 \ \forall u \in U\}.$$

If we apply $\perp$ twice, we get

$$(U^{\perp})^{\perp} = U.$$

Property

$$T \colon V \to W$$

1. $\ker T^* = (\operatorname{im} T)^{\perp}$
2. $\operatorname{im} T = (\ker T^*)^{\perp}$
3. $\ker T = (\operatorname{im} T^*)^{\perp}$
4. $\operatorname{im} T^* = (\ker T)^{\perp}$

Proof. These relationships can be formed from each other

- (1) $\implies$ (2) by taking orthogonal complements ($\perp$)
- (1) $\implies$ (3) $T \mapsto T^*$
- (2) $\implies$ (4)

So we only need to prove (1).

$$\ker T^* = (\operatorname{im} T)^{\perp}$$

Let $w \in W$, then

$$\begin{aligned} w \in \ker T^* &\iff T^* w = 0 \\ &\iff \langle T^* w, v \rangle = 0 \quad \forall v \in V \\ &\iff \langle w, Tv \rangle = 0 \quad \forall v \in V \\ &\iff w \in (\operatorname{im} T)^{\perp}. \end{aligned}$$

□

Definition 20.3

Recall orthonormal basis

Let $\{e_i\}$ be an orthonormal basis of V. Then for any $v \in V$,

$$v = \sum_i \langle v, e_i \rangle e_i.$$

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Property

For linear transformations,

$$T: V \to W.$$

Choose two orthonormal bases,

$$\begin{aligned} e_1, e_2, \ldots, e_n &\quad \text{Orthonormal basis of } V \\ f_1, f_2, \ldots, f_m &\quad \text{Orthonormal basis of } W \end{aligned}$$

Then we have

$$M(T^*) = (M(T))^* = \overline{M(T)^t}$$

Remark

This theorem only works for the matrix corresponding to the chosen *orthonormal bases*.

Proof. We have

$$\begin{aligned} Te_i &= \sum_j f_j M(T)_{ji} = \sum_j \langle Te_i, f_j \rangle f_j, \\ T^* f_j &= \sum_i e_i M(T^*)_{ij} = \sum_i \langle T^* f_j, e_i \rangle e_i. \end{aligned}$$

Compare the coefficients of f_j on both sides,

$$\begin{aligned} M(T)_{ji} &= \langle Te_i, f_j \rangle \\ M(T^*)_{ij} &= \langle T^* f_j, e_i \rangle \end{aligned}$$

Then

$$\langle Te_i, f_j \rangle = \langle e_i, T^* f_j \rangle,$$

which implies

$$M(T)_{ji} = M(T^*)_{ij}.$$

□

Definition 20.4

Consider a linear operator

$$T: V \to V.$$

We call T *self-adjoint* if

$$T = T^*.$$

Remark

A self-adjoint operator is equal to its own adjoint, meaning it satisfies

$$\langle Tv, w\rangle = \langle v, Tw\rangle \quad \forall v, w \in V.$$

Correspondingly, the matrix representation of a self-adjoint operator with respect to an orthonormal basis is equal to its own conjugate transpose.

$$M(T) = M(T)^* = \overline{M(T)^t}$$

- For $\mathbb{F} = \mathbb{R}$, the matrix representation of a self-adjoint operator is symmetric.

$$M(T) = M(T)^t$$

- For $\mathbb{F} = \mathbb{C}$, the matrix representation of a self-adjoint operator is Hermitian.

$$M(T) = M(T)^* = \overline{M(T)^t}$$

Property

Eigenvalues of self-adjoint operator $T\colon V \to V$ are real.

Definition 20.5

Recall eigenvalues of a linear operator

$T\colon V \to V$ are scalars $\lambda \in \mathbb{F}$ such that there exists a nonzero vector $v \in V$ with

$$Tv = \lambda v.$$

Proof. First the definition of eigenvalues states that v is an eigenvector of T corresponding to the eigenvalue λ if

$$Tv = \lambda v.$$

Then take the inner product of both sides with v

$$\langle Tv, v\rangle = \langle \lambda v, v\rangle = \lambda\langle v, v\rangle.$$

and

$$\langle v, Tv\rangle = \langle v, \lambda v\rangle = \overline{\lambda}\langle v, v\rangle.$$

Comparing the two expressions, we get

$$\lambda\langle v, v\rangle = \overline{\lambda}\langle v, v\rangle \implies \lambda = \overline{\lambda} \in \mathbb{R}.$$

□

Property

Let $T\colon V \to V$ be a linear operator such that

$$\langle Tv, v\rangle = 0 \quad \forall v \in V.$$

Then

$$\langle Tv, v\rangle = 0 \quad \forall v \in V \implies T = 0.$$

Remark

Note that the property above holds only for complex vector spaces.

Proof. For any vector $u, w \in V$,

$$\begin{aligned} 4\langle Tu, w\rangle = &+ \langle T(u+w), u+w\rangle \\ &+ i\langle T(u+iw), u+iw\rangle \\ &- \langle T(u-w), u-w\rangle \\ &- i\langle T(u-iw), u-iw\rangle. \end{aligned}$$

Since

$$\langle Tv, v\rangle = 0 \quad \forall v \in V.$$

each term on the right-hand side is zero,

$$\begin{aligned} \langle T(u+w), u+w\rangle &= 0 \\ \langle T(u+iw), u+iw\rangle &= 0 \\ \langle T(u-w), u-w\rangle &= 0 \\ \langle T(u-iw), u-iw\rangle &= 0. \end{aligned}$$

Therefore,

$$\langle Tu, w\rangle = 0 \implies Tu = 0 \quad \forall u \in V.$$

Hence $T = 0$. □

Property

For a complex vector space V ($\mathbb{F} = \mathbb{C}$),

$$T \colon V \to V.$$

The relationship takes place,

$$\langle Tv, v\rangle \in \mathbb{R} \quad \forall v \in V \iff T = T^*.$$

Proof. For a complex vector space V, vector $v \in V$, we have

$$\begin{aligned} T = T^* &\iff T - T^* = 0 \\ &\iff \langle (T - T^*)v, v\rangle = 0 \qquad \forall v \\ &\iff \langle Tv, v\rangle - \langle T^*v, v\rangle = 0 \quad \forall v \\ &\iff \langle Tv, v\rangle - \langle v, Tv\rangle = 0 \quad \forall v \\ &\iff \langle Tv, v\rangle = \overline{\langle Tv, v\rangle} \qquad \forall v. \end{aligned}$$

□

Property

For self-adjoint operator T,

$$T \colon V \to V \quad T = T^*.$$

For a self-adjoint operator T, we have

$$\langle Tv, v\rangle = 0 \quad \forall v \in V \implies T = 0.$$

Remark

This theorem is a general case of the previous property where the operator is self-adjoint and it is true for all complex vector spaces including $\mathbb{R}$.

Since situation $\mathbb{F} = \mathbb{C}$ is already proven, we only need to consider $\mathbb{F} = \mathbb{R}$.

Proof. So for any $v \in V$ $(\mathbb{F} = \mathbb{R})$,

$$4\langle Tv, v\rangle = \langle T(v+w), v+w\rangle - \langle T(v-w), v-w\rangle.$$

By the polarization identity, for any $u, v \in V$,

$$\langle Tu, v\rangle = 0.$$

Hence $T = 0$. □

Definition 20.6

A linear operator $T\colon V \to V$ is called *normal* if it commutes with its adjoint, i.e.,

$$T \circ T^* = T^* \circ T.$$

Remark

Normal operators generalize self-adjoint operators, as every self-adjoint operator is normal.
But the inverse is not necessarily true; not every normal operator is self-adjoint.

Theorem 20.4

For a linear operator

$$T\colon V \to V.$$

We have the following relation,

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \quad \forall v \in V.$$

Proof.

$$\begin{aligned} T \text{ is normal} &\iff TT^* = T^*T \\ &\iff \langle TT^*v, v\rangle = \langle T^*Tv, v\rangle \quad \forall v \in V \\ &\iff \|Tv\|^2 = \|T^*v\|^2 \quad \forall v \in V. \end{aligned}$$

□

Property

Basic properties of normal operators

$$T\colon V \to V \quad T \text{ is normal}$$

1. $\ker(T) = \ker(T^*)$
2. $\operatorname{im}(T) = \operatorname{im}(T^*)$
3. $V = \ker(T) \oplus \operatorname{im}(T)$
4. $T - \lambda I$ is normal for any $\lambda \in \mathbb{C}$
5. $Tv = \lambda v \iff T^*v = \overline{\lambda} v$

Proof. 1. for any $v \in V$ we have

$$\begin{aligned} v \in \ker(T) &\iff Tv = 0 \\ &\iff \langle Tv, Tv \rangle = 0 \\ &\iff \langle T^*v, T^*v \rangle = 0 \\ &\iff T^*v = 0 \\ &\iff v \in \ker(T^*). \end{aligned}$$

2. for any $v \in V$ we have

$$\begin{aligned} v \in \operatorname{im}(T) &\iff \exists u \in V \text{ such that } v = Tu \\ &\iff \exists w \in V \text{ such that } v = T^*w \\ &\iff v \in \operatorname{im}(T^*). \end{aligned}$$

3. Use the rank theorem

$$\dim(\ker(T)) + \dim(\operatorname{im}(T)) = \dim(V).$$

We have

$$\dim(\ker(T^*)) + \dim(\operatorname{im}(T^*)) = \dim(V).$$

And with the previous results, we have

$$V = \ker(T) \oplus \operatorname{im}(T) \implies V = \ker(T^*) \oplus \operatorname{im}(T^*).$$

4. Consider $T - \lambda I$ for any $\lambda \in \mathbb{C}$. We have

$$(T - \lambda I)(T - \lambda I)^* = (T - \lambda I)(T^* - \overline{\lambda} I) = TT^* - \lambda T^* - \overline{\lambda} T + |\lambda|^2 I$$

and

$$(T - \lambda I)^*(T - \lambda I) = (T^* - \overline{\lambda} I)(T - \lambda I) = T^*T - \lambda T^* - \overline{\lambda} T + |\lambda|^2 I.$$

Since T is normal, $TT^* = T^*T$, and thus $T - \lambda I$ is normal.

5. Consider the kernel of $T - \lambda I$ for any $\lambda \in \mathbb{C}$. We have

$$0 \neq \ker(T - \lambda I) \implies 0 \neq \ker(T^* - \overline{\lambda} I).$$

□

21 Lecture 21

Spectrum Theorem and Positivity

Definition 21.1

Recall self adjoint operators and normal operators.

- A self adjoint operator T satisfies $T = T^*$.
- A normal operator T satisfies $TT^* = T^*T$.

21.1 Properties of Normal Operators

Property

For a normal operator

$$T \colon V \to V,$$

The eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof. For arbitrary eigenvectors u and v corresponding to distinct eigenvalues λ and μ, we have

$$Tu = \lambda u, \quad Tv = \mu v, \quad (\lambda \neq \mu).$$

Since T is normal,

$$\begin{aligned}(\lambda - \mu)\langle u, v\rangle &= \lambda\langle u, v\rangle - \mu\langle u, v\rangle \\ &= \langle \lambda u, v\rangle - \langle u, \overline{\mu} v\rangle \\ &= \langle Tu, v\rangle - \langle u, T^* v\rangle = 0.\end{aligned}$$

□

Property

Consider the field of real numbers and a self-adjoint operator T,

$$\mathbb{F} = \mathbb{R}, \quad T \colon V \to V, \quad T^* = T.$$

For real numbers $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is invertible.

Proof. Enough to show $\ker(T^2 + bT + cI) = \{0\}$.
Consider an arbitrary vector $u \in V$,

$$\begin{aligned}\langle (T^2 + bT + cI)u, u\rangle &= \langle T^2 u, u\rangle + b\langle Tu, u\rangle + c\langle u, u\rangle \\ &= \|Tu\|^2 + b\langle Tu, u\rangle + c\|u\|^2.\end{aligned}$$

By the Cauchy-Schwarz inequality

$$|\langle u, v\rangle| \leq \|u\|\|v\|,$$

which implies

$$-|\langle u, v\rangle| \geq -\|u\|\|v\|,$$

and

$$\forall x \in \mathbb{R}, x \geq -|x|,$$

we have

$$\begin{aligned}\|Tu\|^2 + b\langle Tu, u\rangle + c\|u\|^2 &\geq \|Tu\|^2 - |b||\langle Tu, u\rangle| + c\|u\|^2 \\ &\geq \|Tu\|^2 - |b|\|Tu\|\|u\| + c\|u\|^2 \\ &= \left(\|Tu\| - \frac{|b|}{2}\|u\|\right)^2 + \left(c - \frac{b^2}{4}\right)\|u\|^2.\end{aligned}$$

Since $b^2 < 4c$,

$$\left(c - \frac{b^2}{4}\right) > 0.$$

Therefore, the expression is > 0 for $u \neq 0$. □

Theorem 21.1

Recall the Minimal Polynomial theorem

Consider a linear operator T on a finite-dimensional vector space V over a field $\mathbb{F}$.

$$\begin{aligned}p(z) &= a_0 + a_1 z + \cdots + a_m z^m, \\ p(T) &= a_0 I + a_1 T + \cdots + a_m T^m.\end{aligned}$$

$\mu(z)$ is the unique monic polynomial of smallest degree such that

$$p(T) \equiv 0.$$

Property

$$T\colon V \to V, \quad T^* = T$$

The minimal polynomial of T is

$$\mu(z) = (z - \lambda_1)(z - \lambda_2)\cdots(z - \lambda_m),$$

where $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$.

Proof. We use that the roots of $\mu(z)$ are the eigenvalues of T.

$$(\mathbb{F} = \mathbb{C}) \quad \text{FHA} \implies \mu(z) = (z - \lambda_1)\cdots(z - \lambda_m),$$

where λ_i are eigenvalues of T. Since

$$T^* = T,$$

it follows that $\lambda_i \in \mathbb{R}$. □

21.2 Spectrum Theorem

Theorem 21.2

Spectrum Theorem for real vector spaces

$$\mathbb{F} = \mathbb{R}, \quad T: V \to V$$

$$T = T^* \iff T \text{ is diagonalizable w.r.t. an ONB.}$$

Proof. ($\Leftarrow$) Trivial since if T is diagonalizable w.r.t. an ONB, then it is self-adjoint.
($\Rightarrow$) Schur's Theorem implies

$$\exists \text{ ONB } e_1, \ldots, e_n \text{ rel to which } m(T) \text{ is upper triangular.}$$

We can write $m(T)$ as an upper triangular matrix in this form,

$$m(T) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Then $Te_1 = a_{11}e_1$ and $Te_2 = a_{12}e_1 + a_{22}e_2$. Also, T^* is lower triangular,

$$m(T^*) = \begin{pmatrix} \overline{a}_{11} & 0 & \cdots & 0 \\ \overline{a}_{12} & \overline{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a}_{1n} & \overline{a}_{2n} & \cdots & \overline{a}_{nn} \end{pmatrix},$$

so

$$T^*e_1 = \overline{a}_{11}e_1 + \overline{a}_{12}e_2 + \cdots + \overline{a}_{1n}e_n.$$

Since T is normal, it implies

$$\|Te_1\|^2 = \|T^*e_1\|^2.$$

We have

$$\|Te_1\|^2 = |a_{11}|^2,$$

and

$$\|T^*e_1\|^2 = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2.$$

Thus,

$$\begin{aligned} |a_{11}|^2 = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2 &\implies 0 = |a_{12}|^2 + \cdots + |a_{1n}|^2 \\ &\implies a_{12} = \cdots = a_{1n} = 0. \end{aligned}$$

Now we have $Te_2 = a_{22}e_2$, and

$$T^*e_2 = \overline{a}_{22}e_2 + \overline{a}_{23}e_3 + \cdots + \overline{a}_{2n}e_n.$$

By the same approach, we get

$$a_{23} = \cdots = a_{2n} = 0.$$

Continuing in this way, $a_{ij} = 0$ for $i < j$. Thus $m(T)$ is diagonal. □

Theorem 21.3

Spectrum Theorem for complex vector spaces

$$\mathbb{F} = \mathbb{C}, \quad T : V \to V$$

$$T = T^* \iff T \text{ is diagonalizable w.r.t. an ONB.}$$

Proof. The proof is identical to the real case using Schur's Theorem. The diagonal entries will be real because $T = T^*$. □

Theorem 21.4

Schur's Theorem

$\exists$ Orthonormal basis $\{e_1, \ldots, e_n\}$ of V relative to which $m(T)$ is upper triangular.

$$m(T) = \begin{pmatrix} a_{11} & * & \cdots & * \\ 0 & a_{22} & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

$\|Te_1\|^2 = |a_{11}|^2$.

21.3 Positivity

Definition 21.2

A self-adjoint operator $T : V \to V$ is called *positive* if

$$\langle Tv, v \rangle \geq 0 \quad \forall v \in V.$$

Definition 21.3

A *square root* of a positive operator $T : V \to V$ is a positive operator $S : V \to V$ such that

$$S^2 = T.$$

Property

$$T: V \to V$$

TFAE

1. T is positive.
2. $T = T^*$ and all eigenvalues of T are non-negative.
3. There exists an orthonormal basis of V relative to which $m(T)$ is diagonal with non-negative entries.

$$m(T) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}, \quad \lambda_i \geq 0.$$

4. T has a positive square root.
5. T has a self-adjoint square root.
6. $T = R^*R$ for some operator $R: V \to V$.

Theorem 21.5

A positive operator

$$T: V \to V$$

has a unique positive square root.

$$\sqrt{T}: V \to V$$

Corollary 21.6

$$T: V \to V \text{ positive}$$
$$\langle Tv, v \rangle = 0 \implies Tv = 0.$$

Proof.

$$\begin{aligned} \langle Tv, v \rangle &= \langle \sqrt{T}\sqrt{T}v, v \rangle \\ &= \langle \sqrt{T}v, \sqrt{T}v \rangle \\ &= \|\sqrt{T}v\|^2. \end{aligned}$$

$$\langle Tv, v \rangle = 0 \implies \|\sqrt{T}v\|^2 = 0 \implies \sqrt{T}v = 0 \implies Tv = 0.$$

□

A Strike

From: **Leyli Mammadova - lmammado@ed.ac.uk** learn@ed.ac.uk
Subject: MATH080802026-7SV1SEM1 (Linear Algebra 2 (2026-2027)[SEM1]): Industrial action impact
Date: October 16, 2026 at 08:14
To: Undisclosed recipients:;



MATH080802026-7SV1SEM1

Linear Algebra 2 (2026-2027)[SEM1]

New announcement from Leyli Mammadova

Industrial action impact

Dear Linear Algebra 2 students,

I waited until the end of week four to send this message, because I hoped that the dispute would be resolved by now. With deep regret I inform you that the Linear Algebra 2 lectures in weeks five and six will not take place, as I will be on strike. There will also be no workshop materials released.

I wanted to emphasise that going on strike is the last resort for staff: we love teaching and care very much about our students and your education.

However, your learning conditions are also our working conditions! From Monday 28 September, I, alongside many colleagues from UCU Edinburgh across the university, have been on strike as a result of University of Edinburgh senior management's catastrophic cuts, including hidden redundancies and a refusal to rule out compulsory redundancies.

Why is industrial action taking place? As you may have seen, senior management's ongoing financially driven restructuring of the University's Colleges and Schools, on which management refuses to meaningfully consult with UCU, has included a decision to cut 90 programmes in the College of Arts, Humanities and Social Sciences (CAHSS) alone.

Despite being financially unjustified, staff cuts of £92 million have been proposed and are already underway, with the potential for 1800 jobs across the university to be cut.

UCU Edinburgh is resisting management's wanton cultural vandalism and the undermining of our students' education and learning conditions. We insist, instead, that management prioritizes staff job security and our programmes' integrity; staff wellbeing; the provision of essential support services to all students, staff, and visitors; and the future of quality higher education and